

ODSA for Ockenden-2025

 [ODSA_README](#)

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Aim

Here I test the consistency of my ODSA 1D spectral signatures from radar and the 2D landscape classes IFPA (derived from ice physics inversion and applied to the topography product).

Ockenden's paper classifies mesoscale (2-30 km) features into different landscape classes. My beta values are found by fitting a restricted power law to a 250m-50km wavelength band. I am looking to a much broader range (not just mesoscale). I am hoping that this broader scale is complementary, since Ockenden notes:

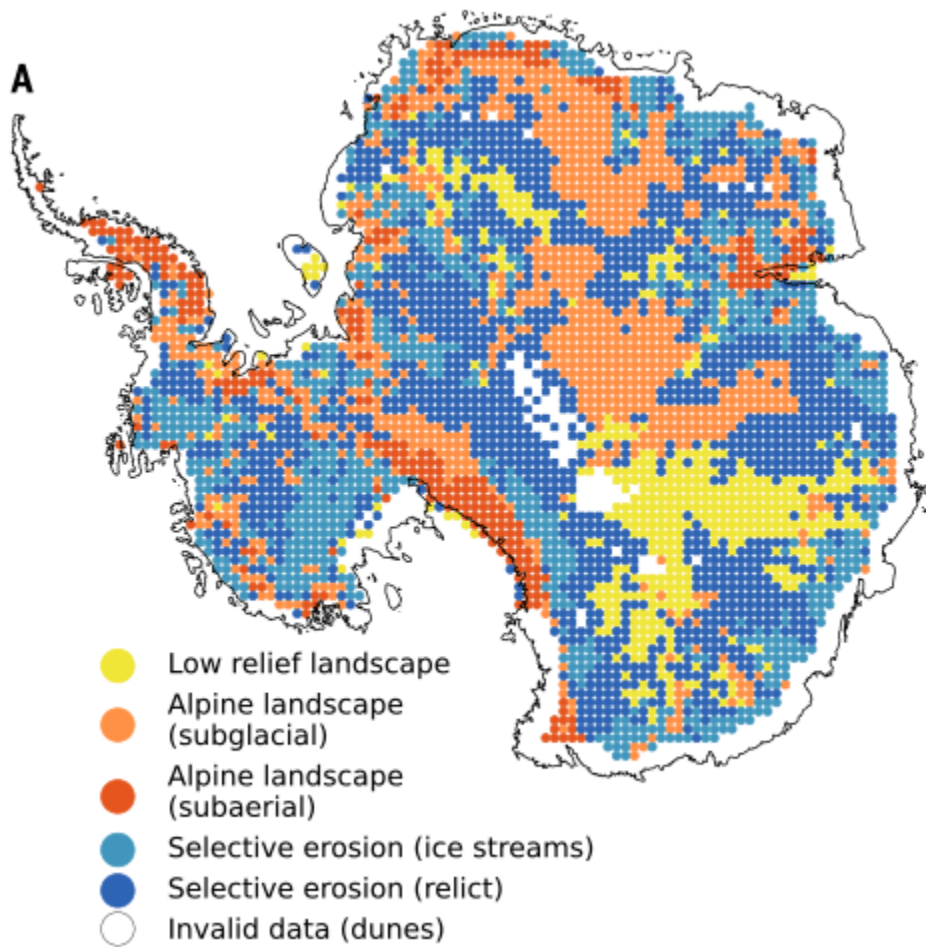
"...subglacial landscapes typically have **fractal roughness structures**. This means the mesoscale textures identified are often correlated with the smaller-scale roughness my code captures".

==IF my spectral analysis from radar data shows high roughness (similarly to IFPA) in a region where Bedmap3 looks smooth , that would be validation for my analysis. ==

SUMMARY

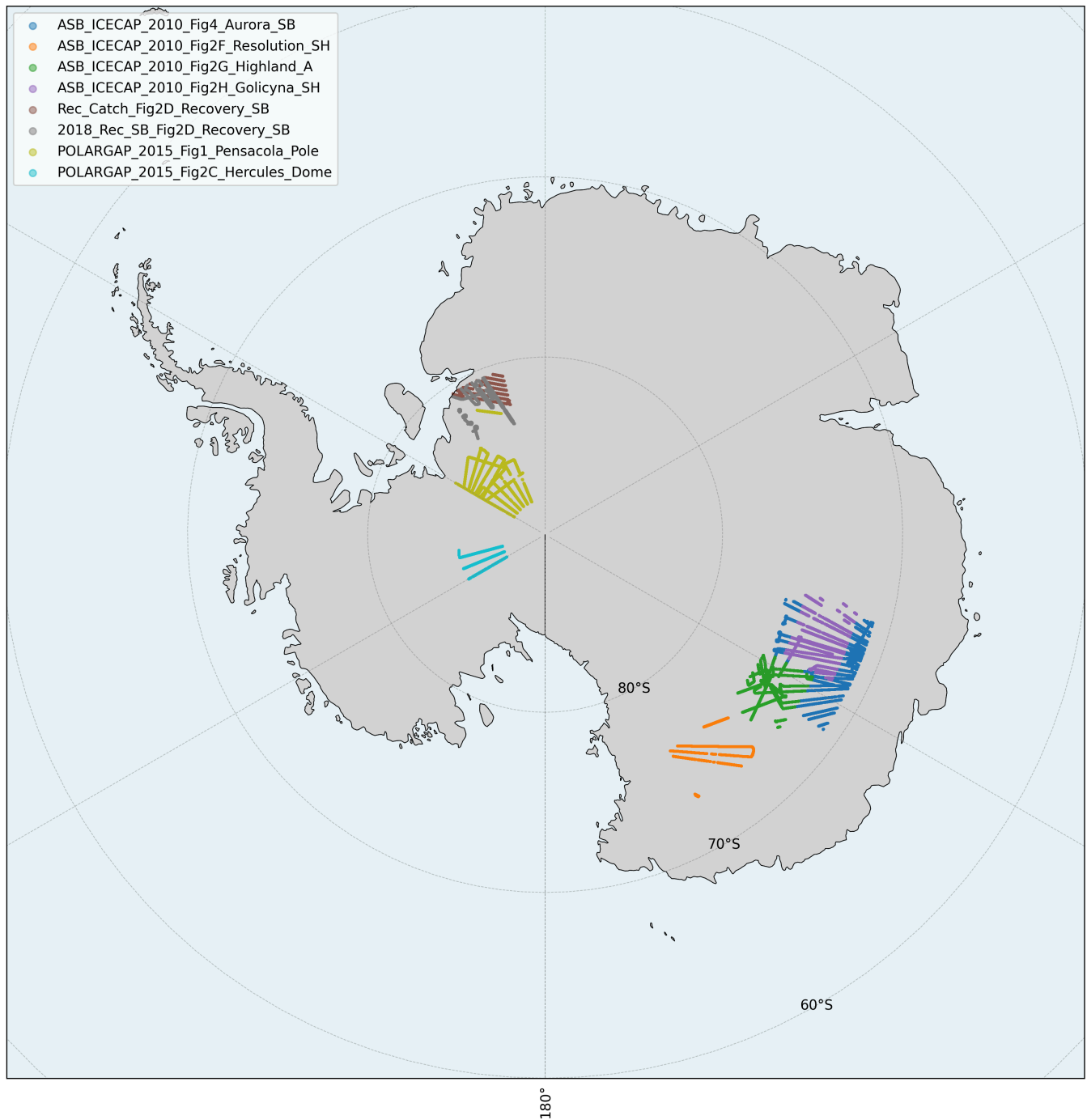
Landscape classification

Ockenden Fig. 4. IFPA Geomorphological classification of Antarctica's subglacial landscape



Selected regions: Radar tracks

Radar Flight Tracks - Antarctica Overview



Landscape classification

Region	IFPA (visual match for radar data tracks)	ODSA	Specific class?
PPB	Alpine/Selective erosion (ice	$\beta = 2.3$, hard bed +	Need further splitting of tracks to draw conclusion

Region	IFPA (visual match for radar data tracks)	ODSA	Specific class?
	streams)	sedimentary	
HD	Alpine/Selective erosion (ice streams)	$\beta = 2.3$, hard bed + sedimentary	Alpine bedrock (high relief) with sedimentary draping ($\beta = 2.3$)
Rec_catch	Low relief / Alpine / Selective erosion (relict)	relief range: [142, 2987] m; mean $\beta = 2.6$, range: [1.6, 3.6] -> Mostly sedimentary	The region contains all three landscape types, but the aggregate signal is dominated by the sedimentary/selective- erosion component.
Rec_2018_SB	Low relief / Alpine / Selective erosion (relict)	$\beta = 2.6$, sedimentary; $\zeta = 0.8$, super- persistent. No significant anisotropy at either scale	Sedimentary. High β and ζ indicate thick sediment draping that has smoothed the underlying structure. The IceBridge tracks sample the trough interiors, unlike the ICEGRAV tracks (Rec_catch) which sample the higher-relief margins
ASB	Low relief / Selective erosion (relict)	$\beta = 1.8$, hard bed	partial? looks like bed is harder than expected
RSH	Low- Relief / Selective erosion (relict)/ Alpine (subglacial)	$\beta = 1.5$, hard bed	partial? looks like bed is harder than expected
HA	Low- Relief / Selective erosion (relict)/ Alpine (subglacial)	$\beta = 2.0$, hard bed + sedimentary	partial? Transitional; alpine component confirmed, selective erosion not supported

Region	IFPA (visual match for radar data tracks)	ODSA	Specific class?
GSH	Low releif/ Selective erosion	$\beta = 1.8$, hard bed	Partial? Hard bed confirmed but anisotropy is alpine, contradicting selective erosion.

Wavelength analysis

Region	Count	Mean (m)	Range	Candidates (>L/2)
PPB	1420	913.4	[54.2, 33400.8]	5
Hercules Dome	492	554.0	[56.1, 15425.2]	1
Rec_catch	2193	1019.5	[30.4, 19221.1]	7
Rec_2018_SB	970	656.9	[30.4, 25232.8]	1
ASB	7416	531.2	[37.7, 31498.9]	29
RSH	1147	516.5	[42.1, 28827.0]	7
HA	2167	578.6	[41.5, 32613.2]	2
GSH	2336	497.6	[35.9, 31510.2]	13

Elevation analysis

Region	Mean vertical relief (m)	Bed Elevation range (m)	IFPA includes "low- relief"?
PPB	1311.3	[-1874.8, 1225.2]	No
HD	1779.4	[-1014.2, 1833.9]	No
Rec_catch	1163.4	[-2325.1, 1865.2]	Yes
Rec_2018_SB	1607.6	[-2192.4, 1689.1]	Yes
ASB	805.3	[-1771.2, 930.4]	Yes
RSH	692.2	[-1254.2, 792.7]	Yes
HA	936.3	[-3445.5, 930.4]	Yes

Region	Mean vertical relief (m)	Bed Elevation range (m)	IFPA includes "low-relief"?
GSH	699.5	[-1453.6, 634.7]	Yes

TODO:

- A future track-by-track or sub-region analysis, separating trajectories by their β values (e.g., $\beta < 2$ vs $\beta > 2.5$), may reveal directional signals within each landscape subset that are currently averaged out.
- A systematic pattern: The ODSA method finds the hard-bed/alpine component is confirmed but the selective erosion/low-relief component is often not found for most regions. The method appears better at detecting hard-bed alpine signatures than selective erosion ones at these scales.

RESULTS

Pensacola-Pole Basin (PPB)

Ockenden points out that many of the PPB areas were "erroneously recorded as flatter ground" in previous Bedmap surveys because they fell in gaps between radar lines.

 Hypothesis

I should observe **high anisotropy** ($\beta_{\parallel} > \beta_{\perp}$)
OR subglacial alpine landscape (**High Relief**, $\beta \sim (1.5 - 2.0)$ and $\beta_{\parallel} = \beta_{\perp}$)

Terminal

```
=== POLARGAP_2015_Fig1_Pensacola_Pole SUMMARY ===
-----
RESULTS SUMMARY (20 trajectories aggregated)
-----
VERTICAL RELIEF (Max-Min):
-> Mean: 1311.3m | Range: [442.2, 2437.3]m
```

ABSOLUTE BED ELEVATION:

-> Min: -1874.8m | Max: 1225.2m (across all trajectories)

AVG SEGMENT LENGTH:

-> Mean: 114066.9m | Range: [20970.2, 262973.5]m

POWER LAW EXPONENT (Beta):

-> Mean: 2.3 | Range: [1.4, 2.9]

HURST EXPONENT:

-> Mean: 0.7 | Range: [0.2, 1.0]

MEAN ICE THICKNESS:

-> Mean: 2990.6m | Range: [1801.3, 3821.0]m

.....

CONFIRMED WAVELENGTHS (Physically valid < L/2):

-> Count: 1420

-> Mean: 913.4m | Range: [54.2, 33400.8]m

CANDIDATE WAVELENGTHS (Statistically present > L/2):

-> Count: 5

-> Range: [10416m, 16815m]

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Exported 229 window rows and 38 segment rows

Observations

Roughness

The mean $\beta = 2.3$ (range [1.4, 2.9]) sits at the boundary between hard-bed and sedimentary landscapes, indicating a heterogeneous mix of **hard bedrock** and **sedimentary** terrain, suggesting that sedimentary infill in trough floors is raising the mean. The Hurst exponent $\zeta = 0.7$ (range [0.2, 1.0]) indicates predominantly persistent topography, where elevation trends continue over distance rather than reversing.

Combining β and ζ points to roughness dominated by long-wavelength undulations.

Relief and bed elevation

The vertical relief is 1311.3 m (range: [442.2, 2437.3]m) is high, consistent with the Alpine/Selective erosion and inconsistent with a low-relief landscape. The total variation in relief across trajectories reflects substantial heterogeneity (likely from tracks crossing both deeply incised troughs (relief > 2000 m) and more subdued terrain (relief ~ 440 m)).

The bed spans from -1874.8 m to 1225.2 m, a total elevation range of ~3100 m.

The deepest points are well below sea level, indicating a marine-influenced ice sheet base in the troughs.

Wavelength structure

The mean confirmed wavelength of 913.4 m (range [54 m, 33.4 km]) indicates the **roughness is dominated by structures at roughly the sub-km-scale.**

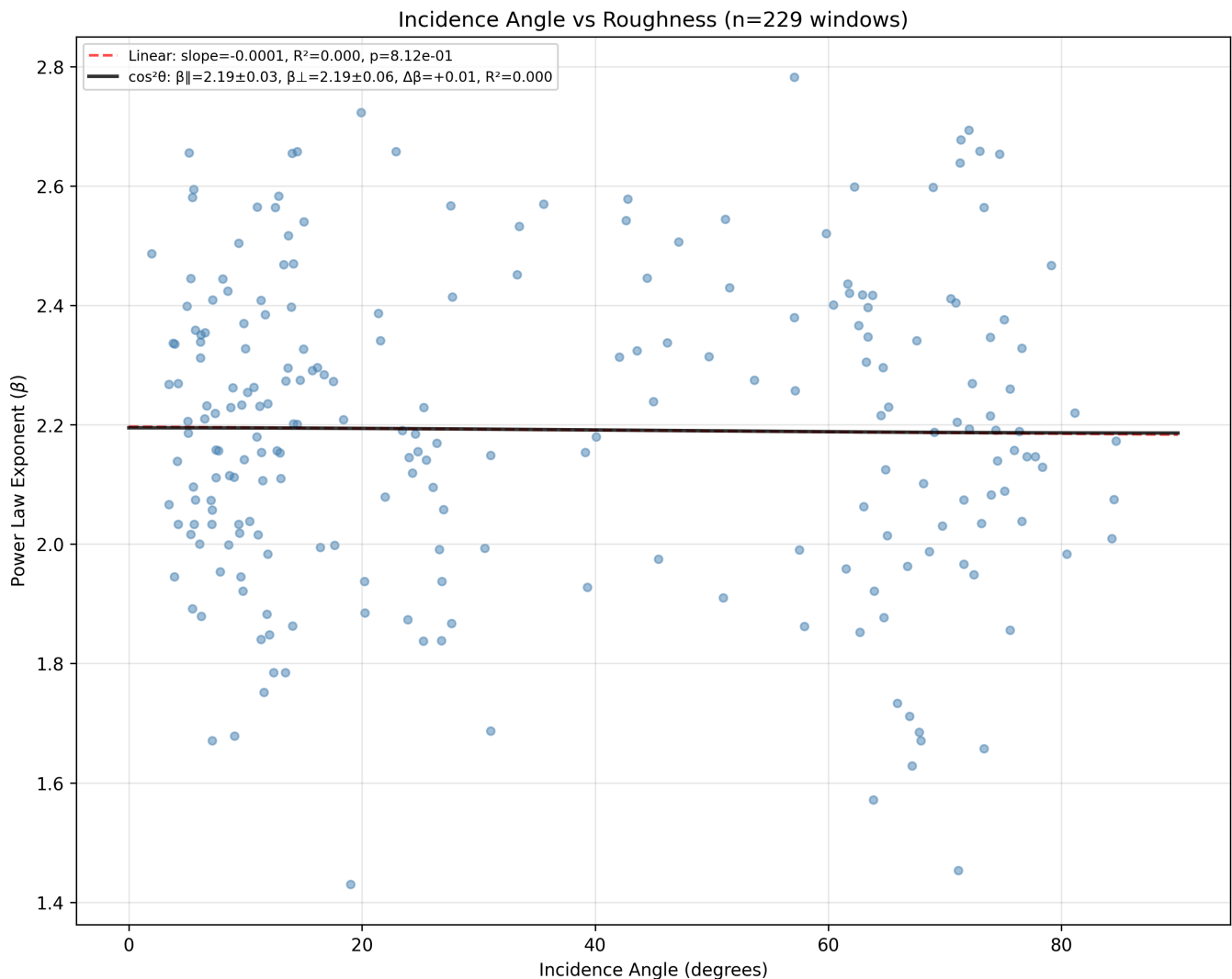
Five candidate wavelengths exceeding $L/2$ are detected, clustered in a narrow band (10.4–16.8 km).

The mean segment length of ~114 km (range [21, 263] km) provides adequate resolution for detecting structures at these scales.

Mean ice thickness is 2990.6 m (range [1801, 3821] m).

Anisotropy

Window-level Roughness FIT



The window level analysis shows a linear fit with slope=-0.0001, $R^2=0.000$, $p=8.12e-$

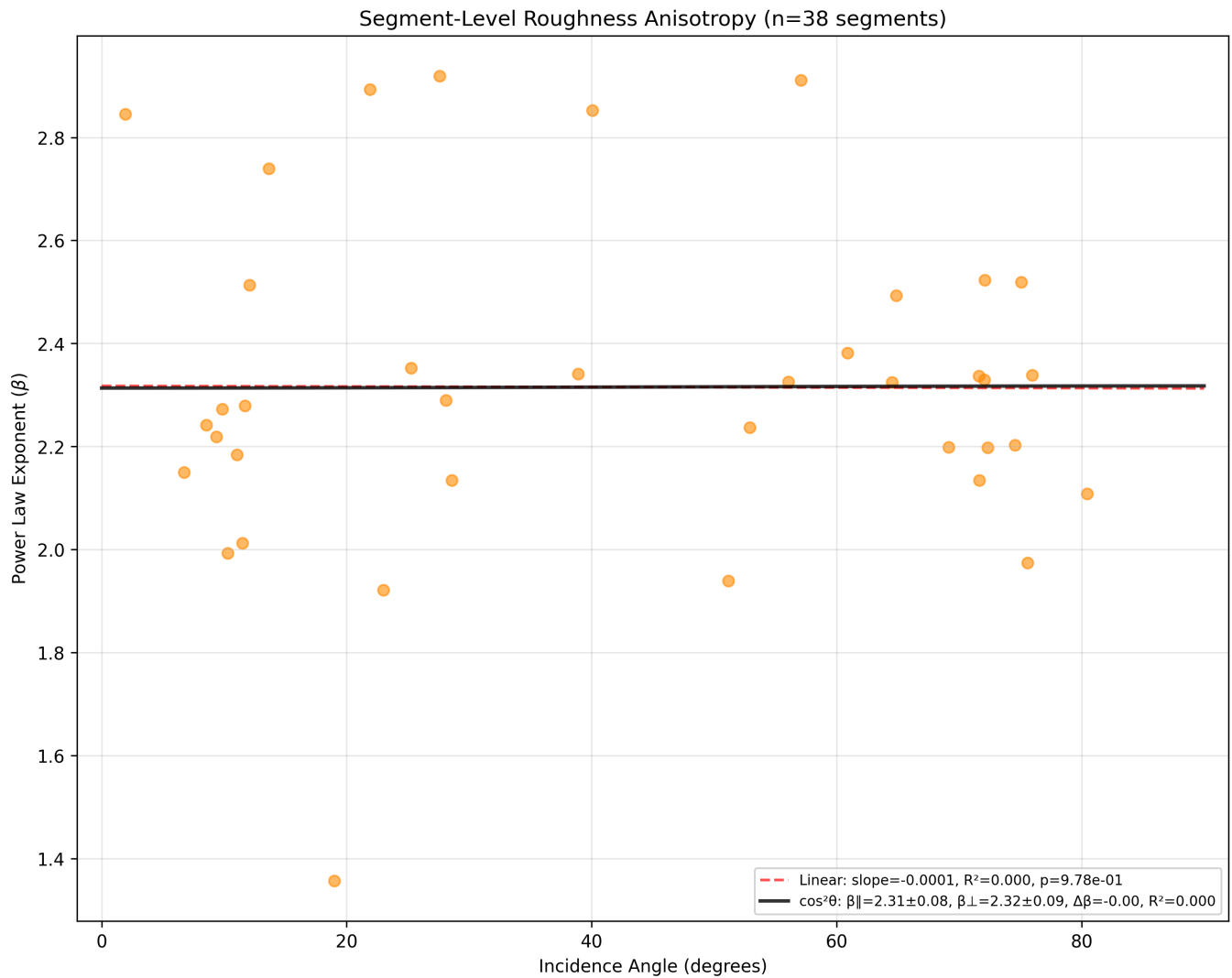
01

At $p=0.812$ this result is non-significant, which implies that the slope is essentially zero.

The $\cos^2 \theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = +0.009$, which provides very weak support for the **Selective erosion landscape**. However the the fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.009}{\sqrt{0.030^2 + 0.057^2}} \approx \frac{0.009}{0.064} \approx 0.140$

This is below 1.96 (95% confidence interval), which means that the result is not significant. Since the dataset is small and the $R^2=0.0002$ implies that the model explains only 0.02% of variance. **No directional dependence detected.**

Segment-level Roughness FIT



The segment level analysis shows a linear fit with -0.0001, $R^2=0.000$, $p=9.78e-01$. At $p=0.978$ this result is non-significant.

The $\cos^2 \theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = -0.004$, which provides support for the **selective erosion landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.004}{\sqrt{0.082^2 + 0.091^2}} \approx \frac{0.004}{0.123} \approx 0.033$

This is below 1.96 (95% confidence interval), which means that the result is not significant. $R^2=0.0000$ implies that the model explains none of the variance. This confirms the window-level result: **no detectable anisotropy at either scale**.

Conclusion:

1. The confirmed wavelength mean is notably higher than the ICECAP regions (ASB: 531 m, RSH: 517 m, GSH: 498 m), suggesting the dominant roughness features are spaced more widely, consistent with broader trough/ridge structures rather than fine-scale crystalline bedrock roughness.
2. Given the wide β range [1.4, 2.9] and high relief variability, the null anisotropy result (at both window and segment levels) is more likely explained by landscape heterogeneity masking any underlying signal than by true isotropy (different landscape types within the region cancel each other's directional signatures).

Hercules Dome (HD)

Alpine (subglacial)/Selective erosion

Hypothesis

I should observe subglacial alpine landscape characteristics:
High Relief, $\beta \sim (1.5 - 2.0)$ and **isotropy** ($\beta_{\parallel} = \beta_{\perp}$)

Terminal

```
=== POLARGAP_2015_Fig2C_Hercules_Dome SUMMARY ===
-----
RESULTS SUMMARY (4 trajectories aggregated)
-----
VERTICAL RELIEF (Max-Min):
-> Mean: 1779.4m | Range: [996.5, 2848.1]m
```

ABSOLUTE BED ELEVATION:

-> Min: -1014.2m | Max: 1833.9m (across all trajectories)

AVG SEGMENT LENGTH:

-> Mean: 107269.6m | Range: [13922.6, 262626.1]m

POWER LAW EXPONENT (Beta):

-> Mean: 2.3 | Range: [1.9, 2.9]

HURST EXPONENT:

-> Mean: 0.6 | Range: [0.5, 0.9]

MEAN ICE THICKNESS:

-> Mean: 3271.2m | Range: [2934.6, 3844.6]m

.....

CONFIRMED WAVELENGTHS (Physically valid $< L/2$):

-> Count: 492

-> Mean: 554.0m | Range: [56.1, 15425.2]m

CANDIDATE WAVELENGTHS (Statistically present $> L/2$):

-> Count: 1

-> Range: [6415m, 6415m]

.....
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Exported 50 window rows and 12 segment rows

Observations

Roughness

The mean β value of 2.3 with a range: [1.9, 2.9] indicates a mix **hard bed** landscapes with **sedimentary** infill. The mean is above the expected range for a purely alpine landscape ($\beta \approx 1.5\text{--}2.0$), but the lower bound of 1.9 is close. This could reflect alpine bedrock with partial sedimentary draping. All trajectories have $\beta > 1.9$. I.E. None show the very rough hard-bed signatures ($\beta < 1.5$).

The Hurst exponent $\zeta = 0.6$ (range [0.5, 0.9]) is moderately persistent, suggesting smooth, trending topography. This landscape is dominated by coherent large-scale structures rather than random roughness.

Relief and bed elevation

The mean vertical relief of 1779.4 m (range: [996.5, 2848.1] m).

Even the lowest-relief (997 m) is close to +1 km, confirming that Hercules Dome sits on consistently rugged terrain. This is strongly consistent with the Alpine (subglacial)/Selective erosion classes and rules out a low-relief landscape.

The height difference between the highest and lowest points available in the data for

this area are Min: -1014.2 m | Max: 1833.9 m (including all available trajectories). The bed reaches substantially above sea level (1833.9 m), suggesting significant subglacial highland terrain. The relatively modest depth below sea level (-1014 m compared to -1875 m for PPB or -2325 m for RSB) suggests less deep marine influence.

Wavelength structure

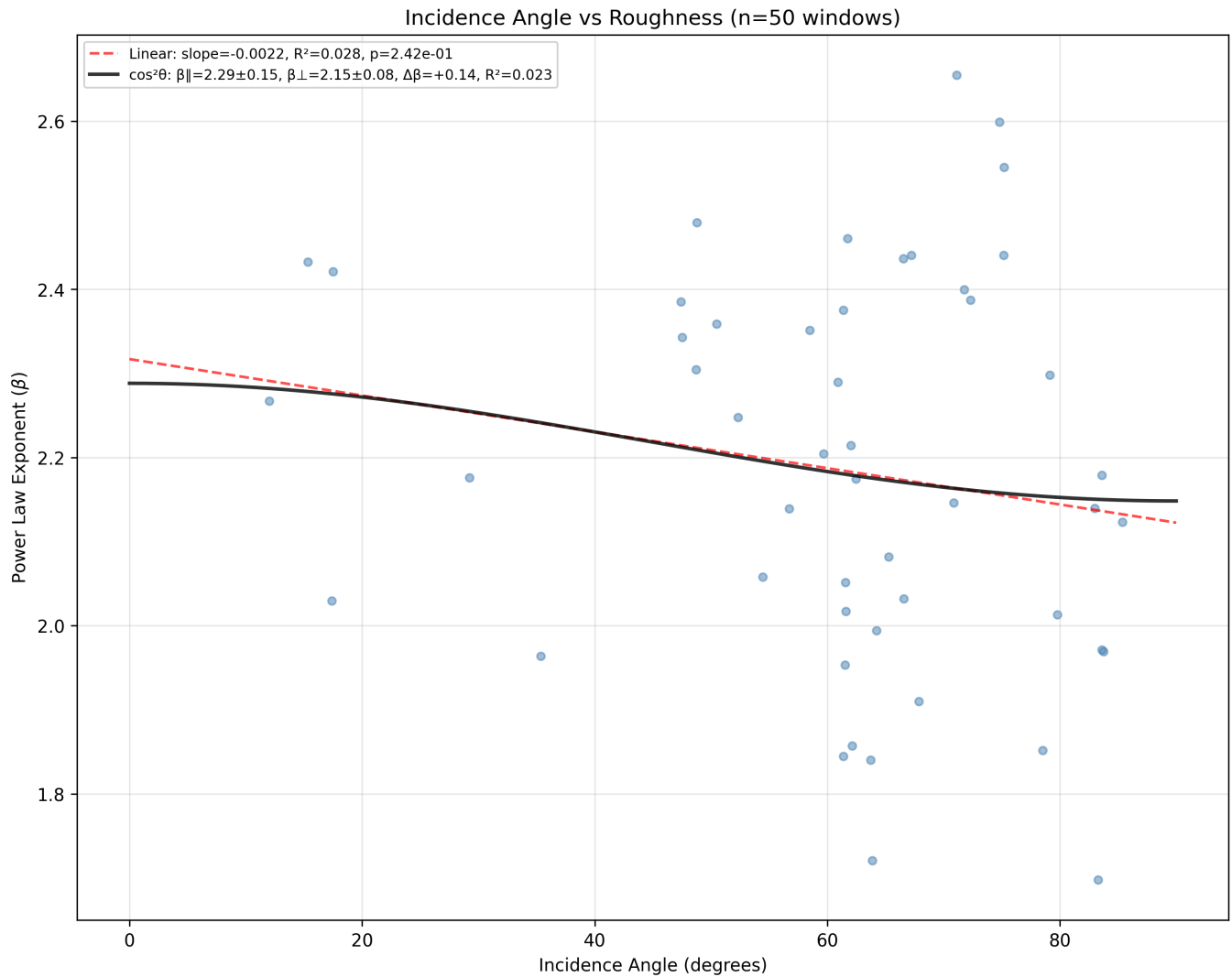
The mean confirmed wavelength of 554.0 m with the range going up to ~15 km indicates the **roughness is dominated by structures at roughly the sub-km-scale**. closely-spaced features consistent with a hard-bed environment where the crystalline bedrock creates high-frequency roughness.

Only 1 candidate wavelength exceeding $L/2$ is detected, at ~6.4 km. The near-absence of large-scale periodic structures contrasts with PPB (5 candidates at 10–17 km) and suggests the terrain lacks regularly spaced troughs or valley systems. This is more consistent with alpine terrain (irregular peaks and valleys) than selective linear erosion (evenly spaced troughs). However, the dataset is small (4 trajectories, 50 windows, 12 segments), limiting the ability to detect large-scale periodicity. The mean segment length of ~107 km (range [14, 263] km) should be adequate for detecting features up to ~50 km, but with only 4 segments the statistical power is low.

Mean ice thickness is 3271.2 m (range [2935, 3845] m).

Anisotropy

Window-level Roughness FIT



The window level analysis shows a linear fit with slope=-0.0022, $R^2=0.028$, $p=2.42e-01$

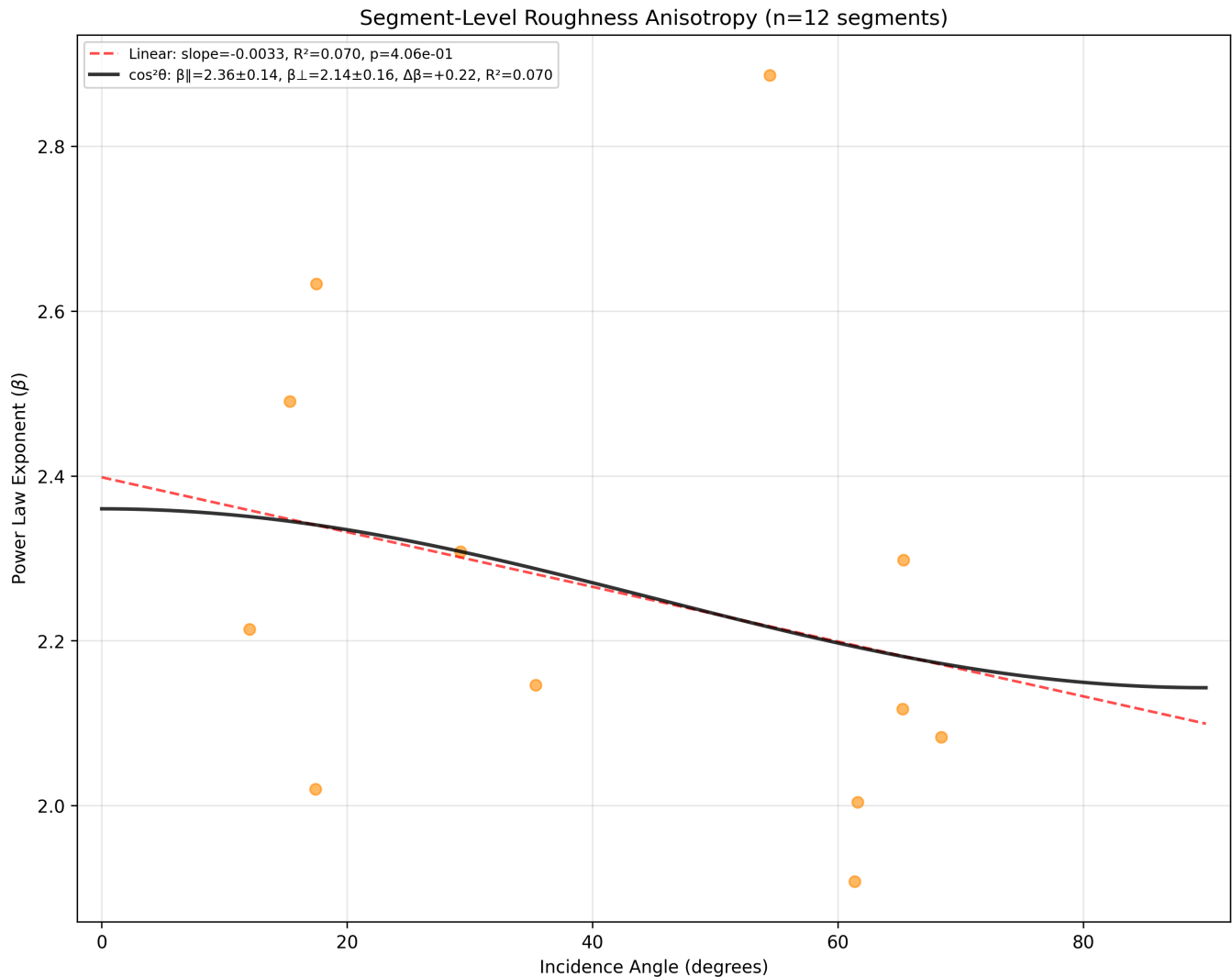
At $p=0.242$ this result is non-significant (there's a 24% chance to see this slope (or steeper) from pure noise). The small sample size (50 windows) limits statistical power (a real but moderate signal could be present but undetectable).

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = +0.140$, which provides support for the **selective erosion landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.140}{\sqrt{0.150^2 + 0.077^2}} \approx \frac{0.140}{0.169} \approx 0.830$

This is below 1.96 (95% confidence interval), which means that the result is not significant. Since the dataset is small and the $R^2=0.0234$ implies that the model explains only 2.3% of variance. This signals **essentially no directional dependence detected by the model**. The result is either uninterpretable at this scale or the signature indicates **Alpine** landscape.

Segment-level Roughness FIT



The segment level analysis shows a linear fit with slope=-0.0033, $R^2=0.070$, $p=4.06e-01$

At $p=0.496$ this result is non-significant (there's a 50% chance you'd see this slope (or steeper) from pure noise).

The $\cos^2 \theta$ anisotropy signal is $\Delta\beta = \beta_{||} - \beta_{\perp} = +0.217$, which provides weak support for the **selective erosion landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{||}^2 + \sigma_{\perp}^2}} = \frac{0.217}{\sqrt{0.144^2 + 0.158^2}} \approx \frac{0.217}{0.214} \approx 1.015$

This is below 1.96 (95% confidence interval), which means that the result is not significant. $R^2=0.0695$ implies that the model explains 7% of variance. In addition the dataset is too small to draw any meaningful conclusion from this analysis.

Conclusion:

1. HD shows the **highest mean relief** (1779 m) of all regions and a **moderately heterogeneous** β range [1.9, 2.9]. The consistently high relief and near-absence of large-scale periodic structures (only 1 candidate wavelength) favour an **alpine landscape** interpretation.
2. The anisotropy analysis shows a **weak positive for both scales** (+0.14 at window, +0.22 at segment), hinting at selective erosion, but neither is **statistically significant**. The hypothesis predicted isotropy for alpine terrain; the observed weak positive signal neither confirms nor refutes this.
3. The $n = 4$ trajectories / 12 segments is really too small to meaningfully constrain anisotropy direction.
4. Mean ice thickness is the second highest after GSH, reflecting the deep interior location of this region.

Recovery Subglacial Basin

Hypothesis

The RSB region should show high-relief alpine/crystalline bed signatures ($\beta \sim 1.5\text{--}2.0$) with substantial vertical relief. Tracks crossing the selective erosion zone near ice streams should exhibit **strong anisotropy** ($\beta_{\parallel} > \beta_{\perp}$), while tracks over the alpine highlands should show weaker directional dependence.

Terminal

```
=== 2018_Rec_SB_Fig2D_Recovery_SB SUMMARY ===
```

```
-----  
RESULTS SUMMARY (50 trajectories aggregated)  
-----
```

```
VERTICAL RELIEF (Max-Min):
```

```
-> Mean: 1163.4m | Range: [141.9, 2986.9]m
```

```
ABSOLUTE BED ELEVATION:
```

```
-> Min: -2325.1m | Max: 1865.2m (across all trajectories)
```

```
AVG SEGMENT LENGTH:
```

```
-> Mean: 38014.5m | Range: [12267.1, 70742.6]m
```

```
POWER LAW EXPONENT (Beta):
```

```

-> Mean: 2.6 | Range: [1.6, 3.6]
HURST EXPONENT:
-> Mean: 0.8 | Range: [0.3, 1.3]
MEAN ICE THICKNESS:
-> Mean: 2159.0m | Range: [627.1, 3881.5]m
.....
CONFIRMED WAVELENGTHS (Physically valid < L/2):
-> Count: 2193
-> Mean: 1019.5m | Range: [30.4, 19221.1]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
-> Count: 7
-> Range: [6172m, 32938m]
.....
=====
Exported 55 window rows and 53 segment rows

```

Observations

Roughness

The mean $\beta = 2.6$ with a range: [1.6, 3.6]. This result indicates a heterogeneous mix of some **hard bed** landscapes with most landscapes in the **sedimentary** regime. The Hurst exponent $\zeta = 0.8$ (range [0.3, 1.3]), some trajectories indicating **super-persistent (trending) topography ($\zeta > 1$)** (long, smooth slopes characteristic of sedimentary basin fill).

The combination of high β and ζ points to a landscape where **sedimentary processes dominate**: the bed is smooth at short wavelengths with power concentrated at long wavelengths, consistent with thick sediment draping over underlying structure.

Relief and bed elevation

The mean vertical relief of 1163.4 m (range [142, 2987] m) shows a huge variation in elevation: Some trajectories cross nearly flat terrain (142 m relief) while others traverse enormous topographic contrasts (nearly 3 km). This is consistent with the IFPA since the region genuinely contains all "Low relief / Alpine / Selective erosion (relict)" landscapes.

The bed spans from -2325.1 m to 1865.2 m. The deep negative values indicate major below-sea-level troughs, while elevations above 1800 m indicate substantial

highlands. This extreme range within one survey region underscores the landscape heterogeneity.

Wavelength structure

The mean confirmed wavelength of 1019.5 m (range [30 m, 19.2 km]) indicates the **roughness is dominated by structures at roughly the km-scale**. These are wide features, which are consistent with a sedimentary environment where the bedrock creates low-frequency roughness and sediment infill smooths out short-wavelength detail. The high wavelength mean also fits with $\beta=2.6$ a sharper power spectrum means more relative power at long wavelengths, so the peak-finding is identifying many large features.

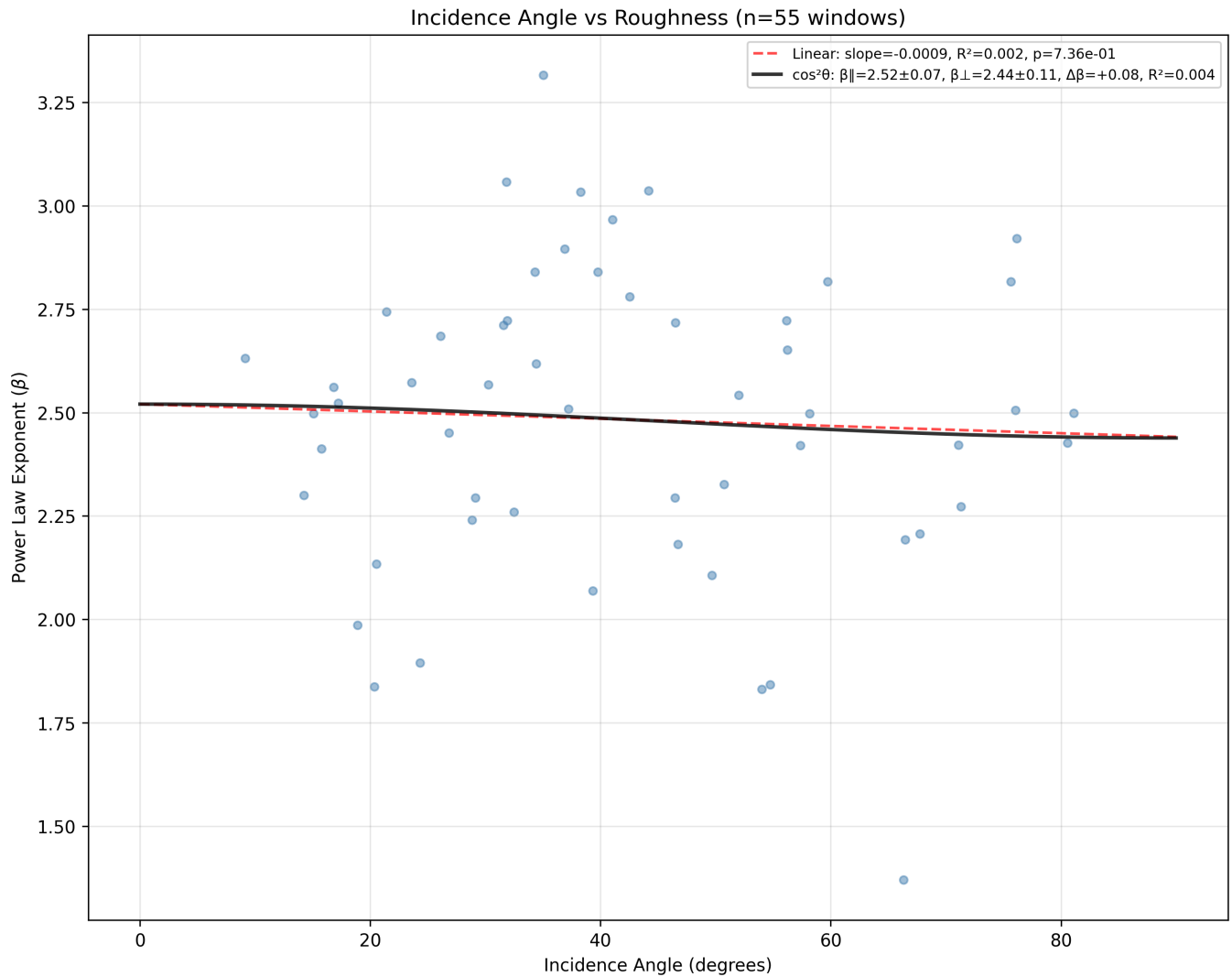
Seven candidate wavelengths exceeding $L/2$ are detected, spanning a broad 6.2–32.9 km range. These candidates are spread across a wide band, suggesting **multiple scales of periodic structure** (possibly nested trough systems at different scales, or superimposed erosion signatures from different ice-flow phases).

The mean segment length of ~38 km (range [12, 71] km) limits the maximum detectable wavelength and may mean even larger periodic structures exist but cannot be resolved.

Mean ice thickness is 2159.0 m (range [627, 3882] m), the wide range and low minimum (627 m) indicate some trajectories pass near the **ice margin** or over shallow ice, potentially over exposed highlands.

Anisotropy

Window-level Roughness FIT



The window level analysis shows a linear fit with slope=-0.0009, $R^2=0.002$, $p=7.36e-01$

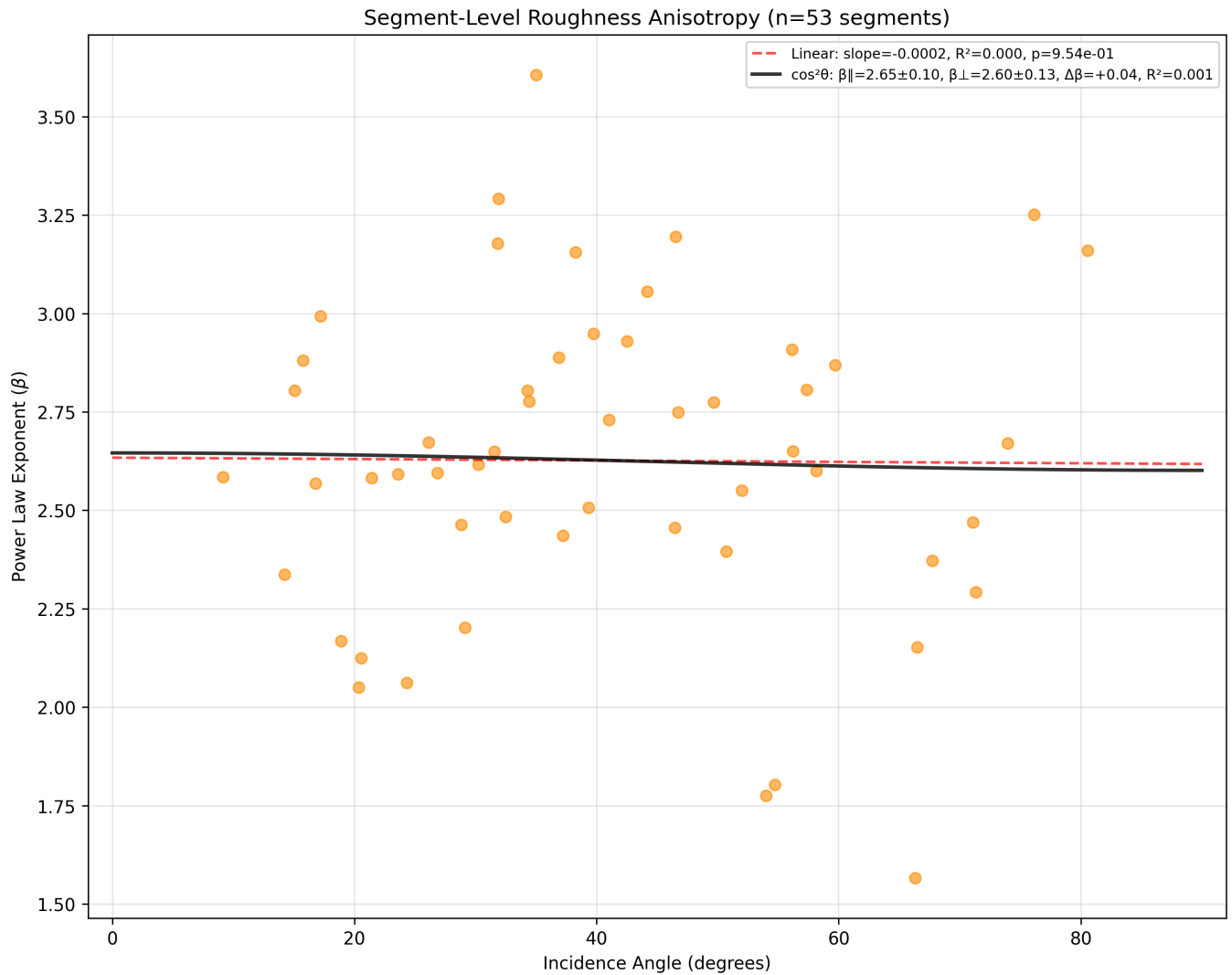
At $p=0.736$ this result is not significant (there's a 74% chance you'd see this slope (or steeper) from pure noise).

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{||} - \beta_{\perp} = +0.082$, which provides support for the **Selective erosion landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{||}^2 + \sigma_{\perp}^2}} = \frac{0.082}{\sqrt{0.075^2 + 0.106^2}} \approx \frac{0.082}{0.130} \approx 0.6315$

This is above 1.96 (95% confidence interval), which means that the result is not significant. The $R^2=0.0042$ value means the directional signal explains 42% of variance.

Segment-level Roughness FIT



The segment level analysis shows a linear fit with slope=-0.0002, $R^2=0.000$, $p=9.54e-01$

At $p=0.954$ this result is non-significant (there's a 95% chance you'd see this slope (or steeper) from pure noise).

The $\cos^2 \theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = +0.044$, which provides support for the **selective erosion landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.044}{\sqrt{0.105^2 + 0.127^2}} \approx \frac{0.044}{0.165} \approx 0.267$

This is below 1.96 (95% confidence interval), which means that the result is not significant. Additionally $R^2=0.0009$ implies that the model explains 0.1% of variance.

Conclusion:

1. Mean β , Hurst exponent, and mean confirmed wavelength all point to a

predominantly sedimentary landscape with large-scale structure.

- 2. This is the only region where the window-level anisotropy is statistically significant.** In addition the larger $\Delta\beta$ at segment scale suggests the directional signal becomes more apparent at longer baselines (~38 km segments), which makes physical sense (trough/ridge anisotropy is a large-scale feature).

Recovery Subglacial Basin: ICEGRAV (RSB-ICEGRAV)

Terminal

```
=== Rec_Catch_Fig2D_Recovery_SB SUMMARY ===
-----
RESULTS SUMMARY (9 trajectories aggregated)
-----
VERTICAL RELIEF (Max-Min):
  -> Mean: 1607.6m | Range: [1036.2, 2905.2]m
ABSOLUTE BED ELEVATION:
  -> Min: -2192.4m | Max: 1689.1m (across all trajectories)
AVG SEGMENT LENGTH:
  -> Mean: 78318.4m | Range: [28829.4, 143748.9]m
POWER LAW EXPONENT (Beta):
  -> Mean: 2.3 | Range: [1.7, 2.8]
HURST EXPONENT:
  -> Mean: 0.7 | Range: [0.3, 0.9]
MEAN ICE THICKNESS:
  -> Mean: 1782.9m | Range: [1004.5, 2624.1]m
.....
CONFIRMED WAVELENGTHS (Physically valid < L/2):
  -> Count: 970
  -> Mean: 656.9m | Range: [30.4, 25232.8]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
  -> Count: 1
  -> Range: [29617m, 29617m]
.....
=====
Exported 84 window rows and 17 segment rows
```

Observations

Roughness

The mean $\beta = 2.3$ (range [1.7, 2.8]) is lower than the 2018 IceBridge dataset for the same region ($\beta = 2.6$), likely reflecting a different subset of trajectories sampling more of the hard-bed terrain. The tighter range (1.1 unit spread vs 2.0 for RSB-2018) suggests these 9 trajectories are more internally consistent. The Hurst exponent $H = 0.7$ (range [0.3, 0.9]) indicates persistent topography, similar to PPB.

Relief and bed elevation

The mean vertical relief of 1607.6 m (range [1036, 2905] m) is substantially higher than RSB-2018 (1163 m), and even the minimum relief (1036 m) exceeds 1 km. This indicates the ICEGRAV trajectories preferentially sample the higher-relief portions of the Recovery Basin, likely the alpine/highland terrain rather than the low-relief sedimentary corridors that dominate RSB-2018.

The bed spans from -2192.4 m to +1689.1 m, a total range of ~3880 m (slightly less extreme than RSB-2018's ~4190 m but still very large).

Wavelength structure

The mean confirmed wavelength of 656.9 m (range [30 m, 25.2 km]) is lower than RSB-2018 (1020 m), consistent with the lower β (a flatter power spectrum means more relative power at short wavelengths, so peak-finding identifies more closely spaced features).

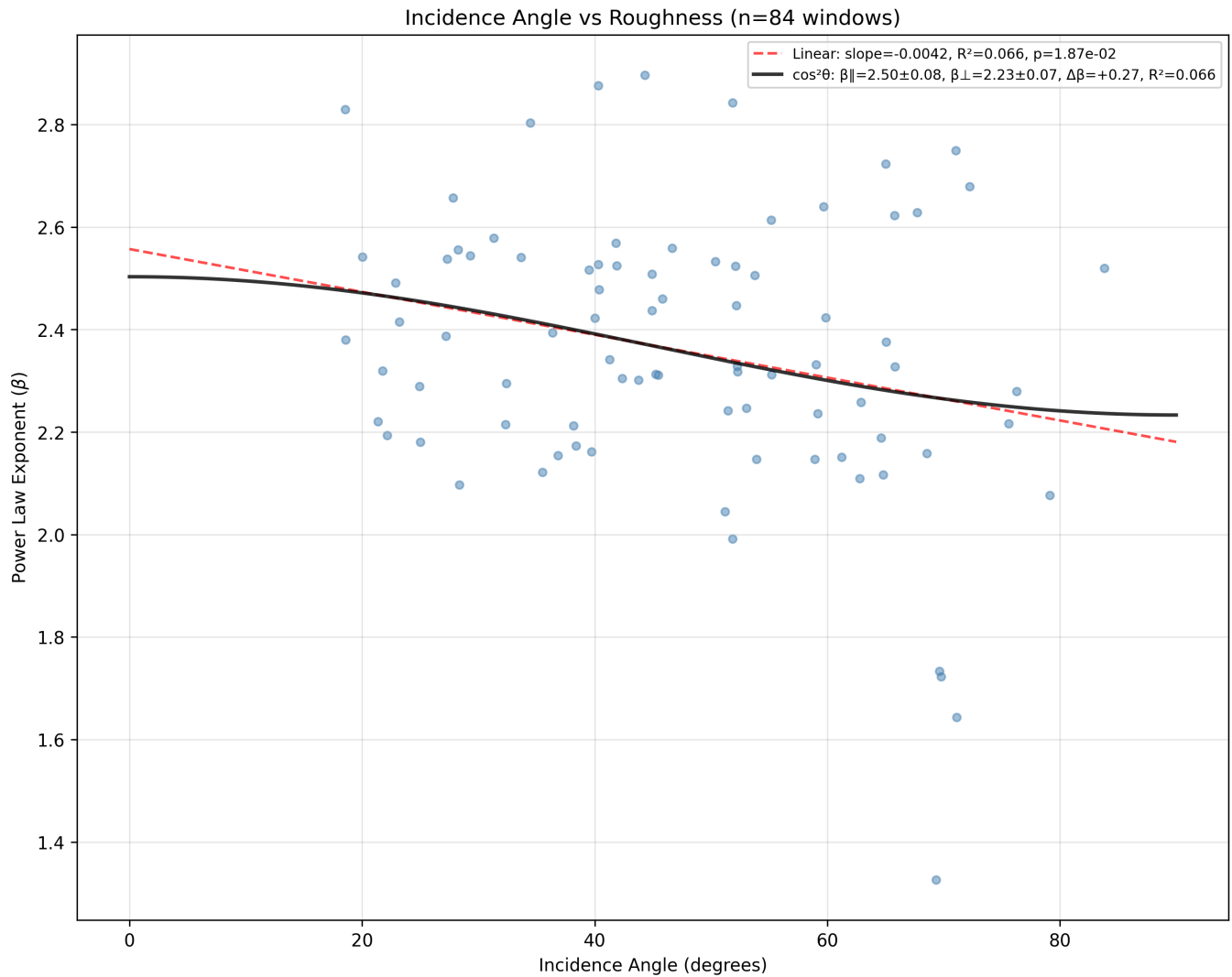
Only 1 candidate wavelength exceeding $L/2$ is detected, at 29.6 km. The contrast with RSB-2018 (7 candidates at 6–33 km) may reflect either the smaller sample size or genuinely different terrain. The ICEGRAV tracks may avoid the regularly spaced trough systems that the IceBridge tracks cross.

The mean segment length of ~78 km (range [29, 144] km) is double that of RSB-2018 (38 km), providing better resolution for large-scale features.

Mean ice thickness is 1782.9 m (range [1005, 2624] m).

Anisotropy

Window-level Roughness FIT



The window level analysis shows a linear fit with slope=-0.0042, $R^2=0.066$, $p=1.87e-02$

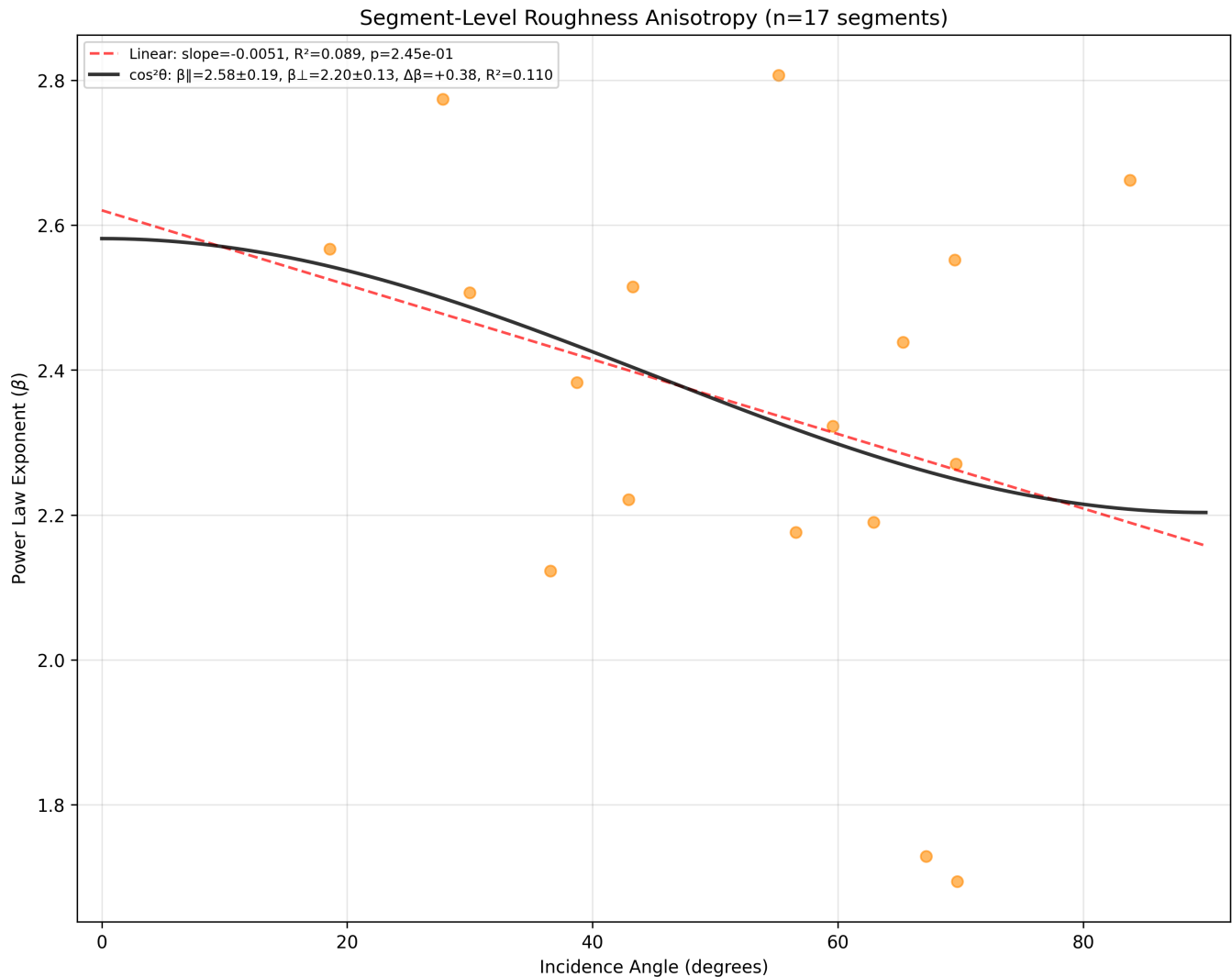
At $p=0.187$ this result is non-significant (there's a 19% chance you'd see this slope (or steeper) from pure noise). The slope is indistinguishable from zero.

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{||} - \beta_{\perp} = +0.270$ is a moderate positive signal of **selective erosion**.

The fit z-score can be calculated by:
$$\frac{|\Delta\beta|}{\sqrt{\sigma_{||}^2 + \sigma_{\perp}^2}} = \frac{0.270}{\sqrt{0.082^2 + 0.074^2}} \approx \frac{0.270}{0.111} \approx 2.445$$

This is above 1.96 (95% confidence interval), which means that the result is significant. Since $R^2=0.0658$ implies that the model explains only 7% of variance.

Segment-level Roughness FIT



The segment level analysis shows a linear fit with slope=-0.0051, $R^2=0.089$, $p=2.45e-01$

At $p=0.245$ this result is non-significant (there's a 25% chance you'd see this slope (or steeper) from pure noise).

The $\cos^2 \theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = +0.378$, A moderate positive which provides support for the **selective erosion landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.378}{\sqrt{0.186^2 + 0.130^2}} \approx \frac{0.378}{0.227} \approx 1.666$

Below 1.96 (95% confidence interval), the result is not significant. $R^2=0.1104$ implies that the model explains 11% of variance.

Conclusion

1. RSB-2018 shows a predominantly sedimentary signature ($\beta = 2.6$, $\zeta = 0.8$, mean wavelength ~ 1020 m) while RSB-ICEGRAV shows a harder-bed, higher-relief

signal ($\beta = 2.3$, relief = 1608 m, mean wavelength ~657 m). Together, these two datasets sample complementary parts of the basin: the IceBridge tracks cross the sediment-filled trough interiors, while the ICEGRAV tracks preferentially sample the highland margins.

2. The RSB hypothesis predicted high-relief alpine/crystalline signatures with strong anisotropy near ice streams and weaker directional dependence over the alpine highlands. The results partially confirm this: the anisotropy signal is real but modest. The high beta values and super-persistent topography ($\zeta > 1$ in some trajectories) suggest thick sediment infill has smoothed the underlying structure, with the directional signal from the trough/ridge geometry persisting through the sediment cover.

UTIG_2010_ICECAP_AIR_BM3

Aurora Subglacial Basin,
Resolution Subglacial Highlands,
Highland A,
Golicyna Subglacial Highlands.

Hypothesis

- Radar tracks for ASB are located in a mixed IFPA region (Low-relief/ Selective erosion (relict)) The expected anisotropy signal is $\beta_{\parallel} > \beta_{\perp}$, also should have **high** β values.
- Important **high relief** regions are to be expected in the Resolution SH, Highland A and Golicyna SH. These areas should have **low** β values and **isotropy**.

Aurora Subglacial Basin

Low Relief / Selective erosion (relict)

Terminal

=== ASB_ICECAP_2010_Fig4_Aurora_SB SUMMARY ===

RESULTS SUMMARY (68 trajectories aggregated)

VERTICAL RELIEF (Max-Min):

-> Mean: 805.3m | Range: [175.2, 2204.8]m

ABSOLUTE BED ELEVATION:

-> Min: -1771.2m | Max: 930.4m (across all trajectories)

AVG SEGMENT LENGTH:

-> Mean: 108781.8m | Range: [13839.3, 542478.1]m

POWER LAW EXPONENT (Beta):

-> Mean: 1.8 | Range: [0.9, 3.1]

HURST EXPONENT:

-> Mean: 0.4 | Range: [-0.1, 1.0]

MEAN ICE THICKNESS:

-> Mean: 3477.1m | Range: [2586.4, 4243.5]m

.....
CONFIRMED WAVELENGTHS (Physically valid < L/2):

-> Count: 7416

-> Mean: 531.2m | Range: [37.7, 31498.9]m

CANDIDATE WAVELENGTHS (Statistically present > L/2):

-> Count: 29

-> Range: [6197m, 41095m]
.....

=====

Exported 827 window rows and 125 segment rows

Observations

Roughness

The mean β value of 1.8 with a range: [0.9, 3.1]. The wide range (2.2 unit spread) is the second-largest observed and spans from very rough crystalline bedrock to smooth sedimentary terrain. The Hurst exponent $\zeta = 0.4$ (range [-0.1, 1.0]) is very low. The values below 0.5 indicate **anti-persistent topography, where elevation tends to reverse direction frequently** (rough, jagged terrain). The minimum $\zeta = -0.1$ means at least one trajectory shows roughness that increases with scale, characteristic of very rough, fractured bedrock. The combination of β and ζ for this region marks the spectral fingerprint of a **hard-bed landscape**: flat power spectra

(equal energy across wavelengths) with anti-persistent profiles (frequent elevation reversals).

Relief and bed elevation

The mean vertical relief of 805.3 m (range [175, 2205] m) is moderate but highly variable. The lower end (175 m) is genuinely low relief, consistent with some IFPA cells being classified as "low relief" in this region. The upper end (2205 m) indicates some trajectories cross major topographic features.

The bed spans from -1771.2 m to +930.4 m. The maximum elevation is modest (+930 m) compared to inland regions like HD (+1834 m), consistent with ASB being a subglacial basin rather than highland terrain.

Wavelength structure

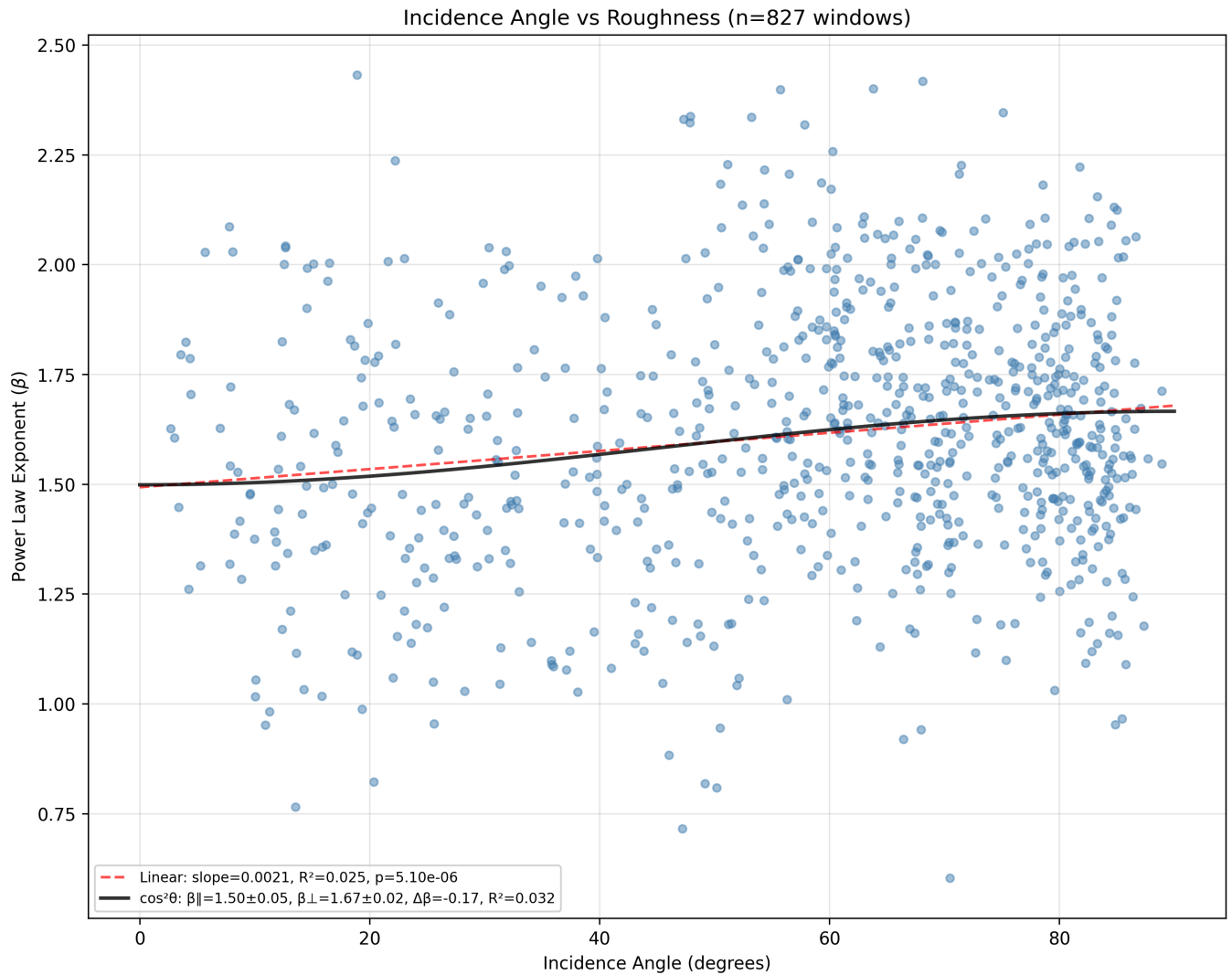
The mean confirmed wavelength of 531.2 m (range [38 m, 31.5 km]) indicates roughness dominated by **sub-km-scale** structures (closely spaced features consistent with crystalline bedrock creating high-frequency roughness). The low wavelength mean fits with $\beta = 1.8$: a flatter power spectrum means more relative power at short wavelengths, so the peak-finding identifies many small features. ASB has 29 candidate wavelengths exceeding $L/2$. These span 6.2–41.1 km. The abundance and spread of large-scale periodic structures suggests **multiple superimposed landform systems** at different scales (possibly subglacial valley/ridge networks with varied orientations and spacings). This could reflect complex erosion history with multiple ice-flow phases.

The mean segment length of ~109 km (range [14, 542] km).

Mean ice thickness is 3477.1 m (range [2586, 4244] m).

Anisotropy

Window-level Roughness FIT



The window level analysis shows a linear fit with slope=0.0021, $R^2=0.025$, $p=5.10e-06$

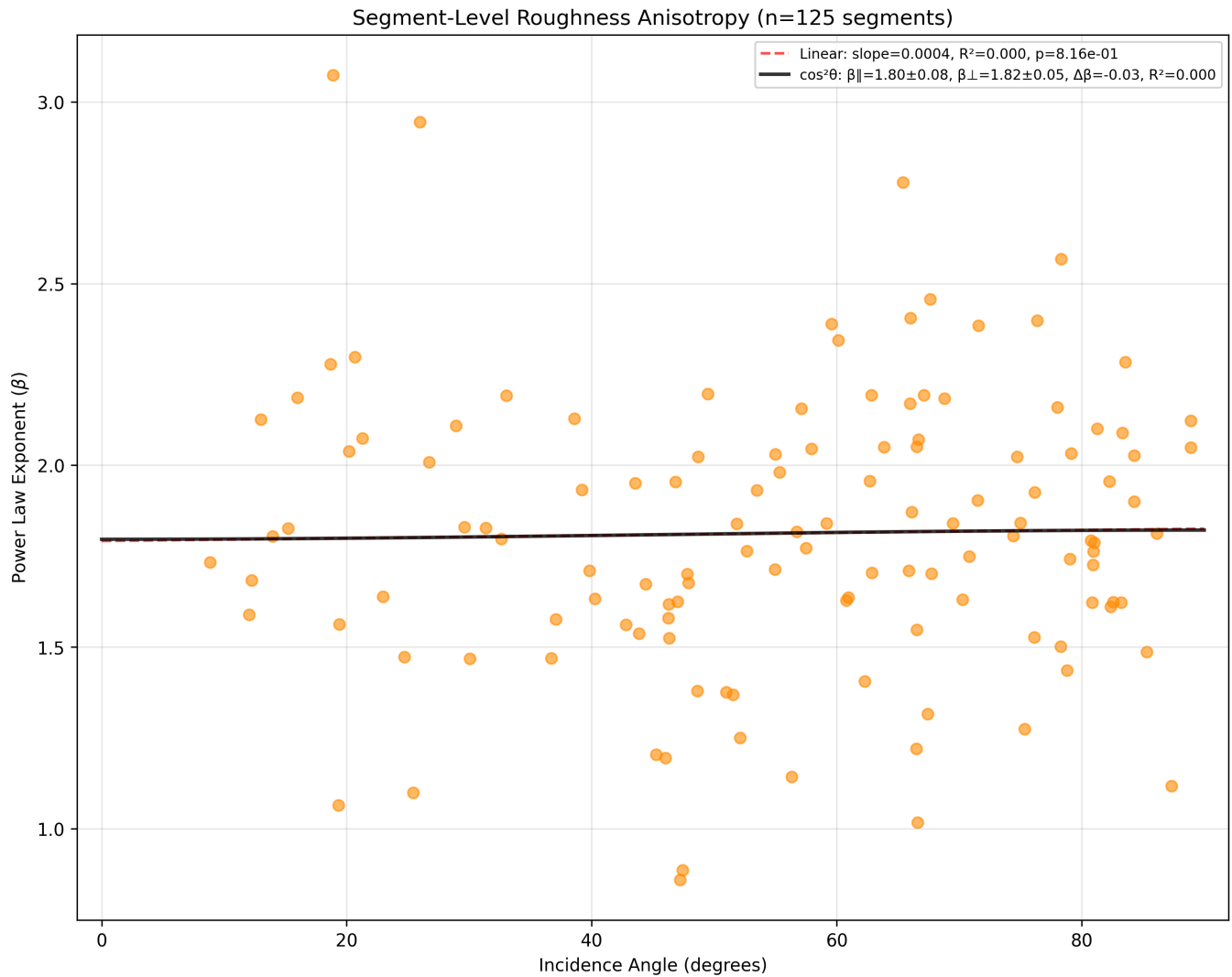
At $p=0.000005$ this result is highly significant. However, $R^2=0.025$ means the directional signal explains only 2.5% of variance.

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = -0.167$, which provides support for the **Alpine landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.167}{\sqrt{0.049^2 + 0.024^2}} \approx \frac{0.167}{0.0546} \approx 3.0607$

This is above 1.96, which means that the result is significant. Nevertheless, the $R^2=0.0316$ implies that the model explains only 3.2% of variance. The landscape is very noisy, and the alpine anisotropy is a subtle effect embedded within high variability.

Segment-level Roughness FIT



The segment level analysis shows a linear fit with slope=0.0004, $R^2=0.000$, $p=8.16e-01$

At $p=0.816$ this result is non-significant.

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = -0.025$, which provides support for the **selective erosion landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.025}{\sqrt{0.082^2 + 0.054^2}} \approx \frac{0.025}{0.098} \approx 0.2546$

This is below 1.96 (95% confidence interval), which means that the result is not significant. While $R^2=0.0004$ implies that the model explains 0.04% of variance.

The anisotropy signal **disappears at the segment level** ($\Delta\beta$ drops from -0.167 to -0.025), and the result is completely non-significant. This sign agreement (both negative) but magnitude collapse suggests the alpine anisotropy is a **mesoscale phenomenon** (detectable within 50 km windows) that averages out over longer

baselines. At segment scale (~109 km), individual trajectories cross multiple ridge/valley systems in different orientations, diluting the directional signal.

Conclusion:

ASB shows **Hard-bed / alpine signature**: low mean β (1.8), low Hurst exponent ($\zeta = 0.4$), and a **highly significant negative** $\Delta\beta$ at window level ($z = 3.06$, $p < 0.00001$). This combination (rough, anti-persistent terrain with β higher perpendicular than parallel to structures) is the spectral fingerprint of an **alpine landscape with valley/ridge anisotropy**.

The IFPA classification of "Low relief / Selective erosion (relict)" is **partially supported**: the low- β hard-bed signature is detected, but the anisotropy is alpine ($\Delta\beta < 0$) rather than selective erosion ($\Delta\beta > 0$). The 29 candidate wavelengths at 6–41 km suggest complex multi-scale structure rather than the simple, regularly spaced troughs expected from selective linear erosion. The moderate mean relief (805 m) with some genuinely low-relief trajectories (175 m) is consistent with the "low relief" component of the IFPA classification.

Resolution Subglacial Highlands

Low- Relief / Selective erosion (relict)/ Alpine (subglacial)

Terminal

```
=== ASB_ICECAP_2010_Fig2F_Resolution_SH SUMMARY ===
-----
RESULTS SUMMARY (7 trajectories aggregated)
-----
VERTICAL RELIEF (Max-Min):
  -> Mean: 692.2m | Range: [413.3, 1034.5]m
ABSOLUTE BED ELEVATION:
  -> Min: -1254.2m | Max: 792.7m (across all trajectories)
AVG SEGMENT LENGTH:
  -> Mean: 87040.0m | Range: [13472.9, 221848.6]m
POWER LAW EXPONENT (Beta):
  -> Mean: 1.5 | Range: [0.8, 2.5]
HURST EXPONENT:
```

```

-> Mean: 0.3 | Range: [-0.1, 0.8]
MEAN ICE THICKNESS:
-> Mean: 3210.5m | Range: [2476.2, 3691.4]m
.....
CONFIRMED WAVELENGTHS (Physically valid < L/2):
-> Count: 1147
-> Mean: 516.5m | Range: [42.1, 28827.0]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
-> Count: 7
-> Range: [8018m, 26048m]
.....
=====
Exported 85 window rows and 22 segment rows

```

Observations

Roughness

The mean β value of 1.5 with a range: [0.8, 2.5]. This result indicates **hard bed** landscape. The minimum indicates extremely rough terrain with a nearly white (flat) power spectrum, characteristic of fractured crystalline bedrock. The Hurst exponent $\zeta = 0.3$ (range [-0.1, 0.8]) is also the lowest, with the negative minimum indicating anti-persistent roughness at the most extreme end.

The low β and low ζ together suggest a landscape where short-wavelength roughness dominates. The bed is jagged and irregular at all scales, with no preferred smoothing at any wavelength.

Relief and bed elevation

The mean vertical relief of 692.2 m (range [413, 1035] m) is moderate and relatively consistent. No trajectory shows either very low (<200 m) or very high (>2000 m) relief. This modest, uniform relief is consistent with a highland plateau that has been moderately dissected but not deeply trenched.

The bed spans from -1254.2 m to +792.7 m. The maximum elevation is relatively low for a "highlands" region compared to HD (+1834 m) suggesting these are subdued highlands rather than dramatic alpine peaks.

Wavelength structure

The mean confirmed wavelength of 516.5 m (range [42 m, 28.8 km]) indicating roughness dominated by closely spaced **sub-km features**, consistent with the hard-bed interpretation. The low wavelength mean fits with the low β (1.5), a flat power

spectrum with equal energy across wavelengths means peak-finding identifies many small features.

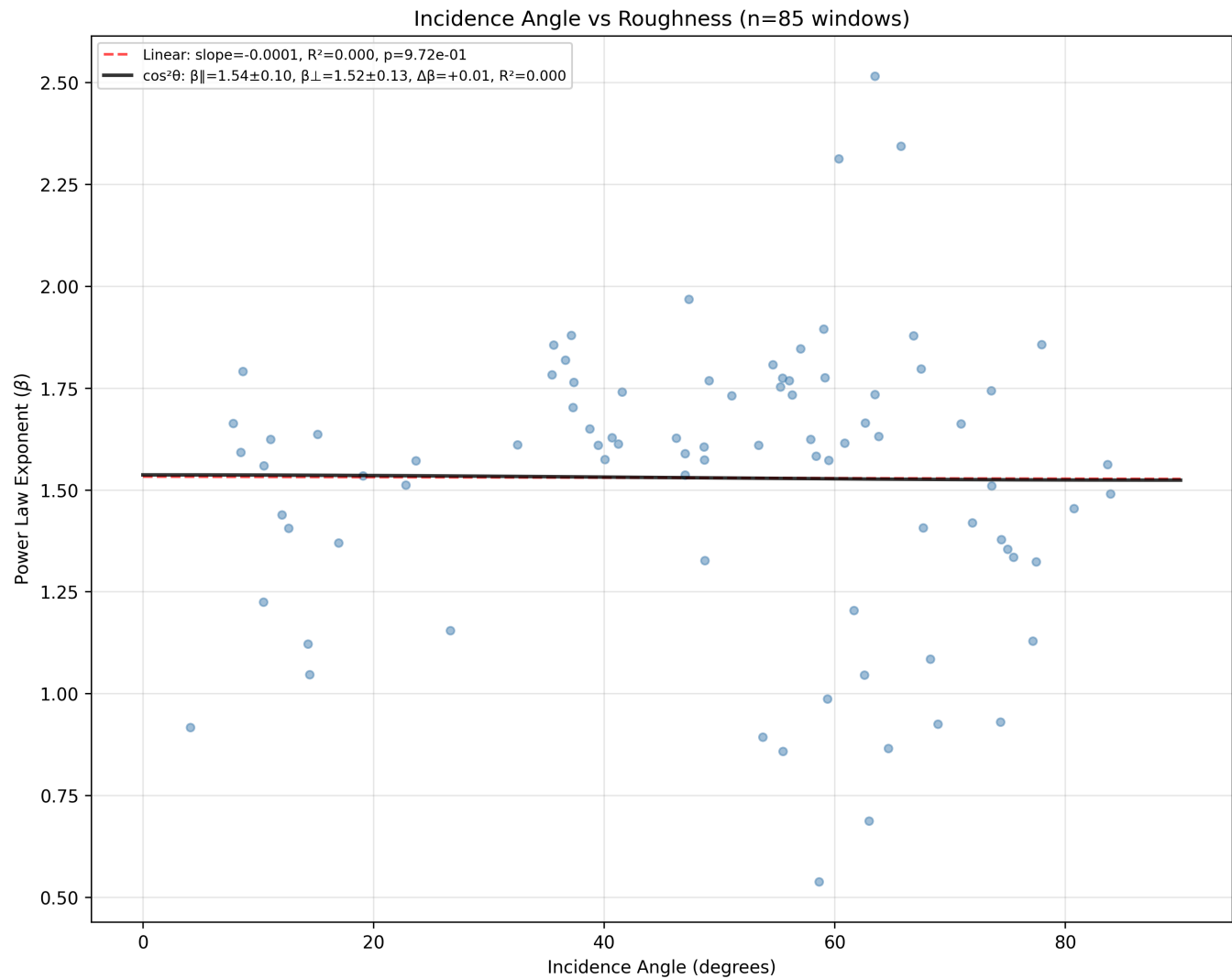
Seven candidate wavelengths exceeding $L/2$ are detected, spanning 8.0–26.0 km. Despite the modest relief, the region does contain large-scale periodic structures, possibly relict valley/ridge systems from pre-glacial drainage patterns.

The mean segment length of ~87 km (range [13, 222] km) is adequate for resolving these scales.

Mean ice thickness is 3210.5 m (range [2476, 3691] m).

Anisotropy

Window-level Roughness FIT



The window level analysis shows a linear fit with slope=-0.0001, $R^2=0.000$, $p=9.72e-01$

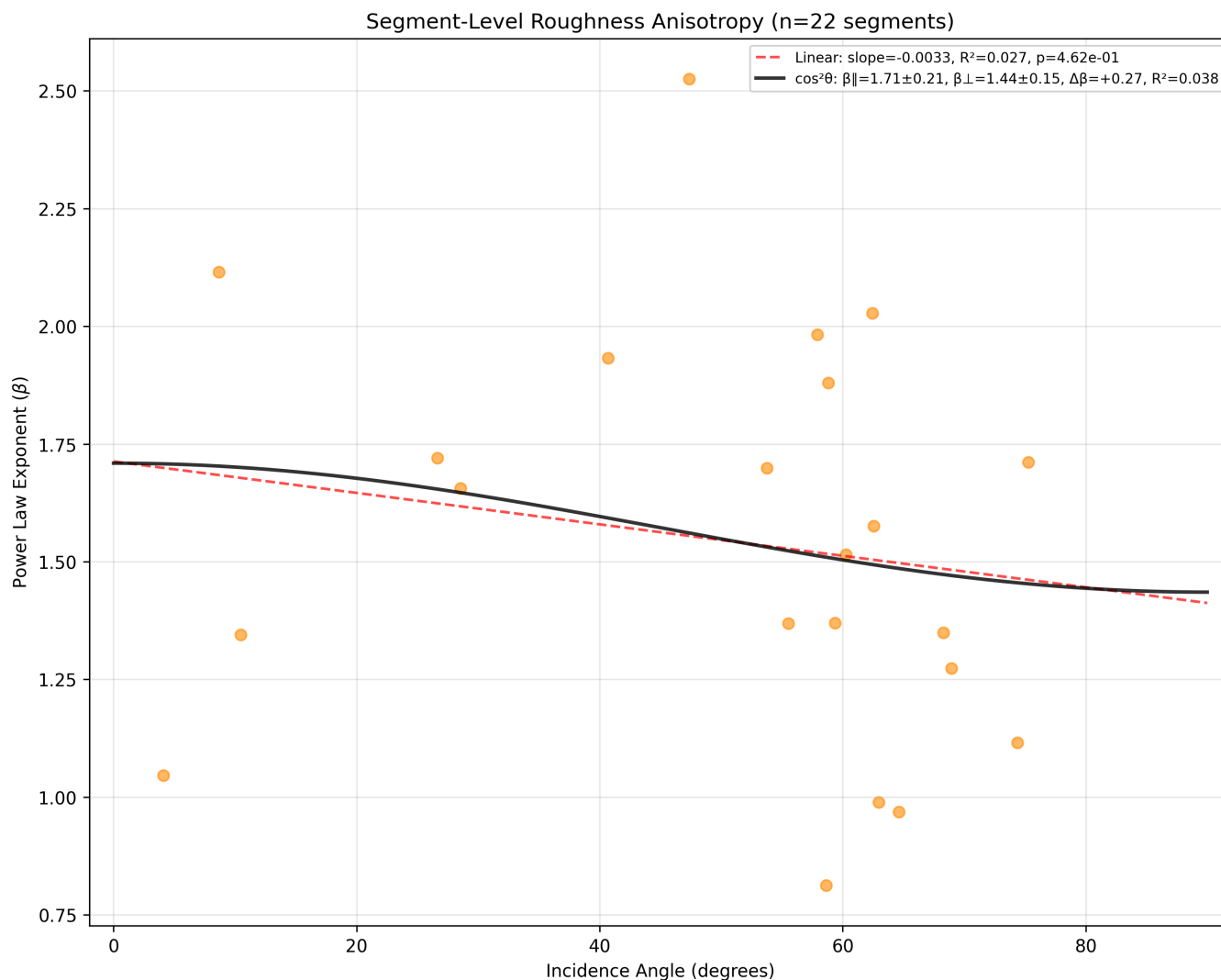
At $p=0.972$ this result is non-significant (there's a 97% chance you'd see this slope (or steeper) from pure noise). The slope is indistinguishable from zero.

The $\cos^2 \theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = +0.013$ is a very very weak signal of **selective erosion**. But could just be considered **isotropic**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.013}{\sqrt{0.102^2 + 0.129^2}} \approx \frac{0.013}{0.165} \approx 0.080$

This is below 1.96 (95% confidence interval), which means that the result is not significant. Since $R^2=0.0001$ implies that the model explains only 0.01% of variance. This means that **no directional dependence detected** at this scale.

Segment-level Roughness FIT



The segment level analysis shows a linear fit with slope=-0.0033, $R^2=0.027$, $p=4.62e-01$

At $p=0.462$ this result is non-significant (there's a 46% chance you'd see this slope (or steeper) from pure noise).

The $\cos^2 \theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = +0.274$, A moderate positive which provides support for the **selective erosion landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.274}{\sqrt{0.208^2 + 0.154^2}} \approx \frac{0.274}{0.259} \approx 1.059$

Below 1.96 (95% confidence interval), the result is not significant. $R^2=0.0377$ implies that the model explains 4% of variance.

Conclusion:

RSH displays an **extreme hard-bed spectral signature**:

β (1.5), ζ (0.3), and sub-km dominant wavelengths. The anisotropy is **isotropic at window level** with a weak, non-significant positive signal at segment level.

The hypothesis predicted low β values and isotropy for highland terrain. The ODSA results **partially support this**: the roughness values clearly indicate hard bed, and the window-level result is isotropic. The moderate, consistent relief (692 m, tight range) is consistent with a highland plateau. However, the IFPA classification includes "Alpine (subglacial)" alongside "Low-Relief / Selective erosion (relict)", while $\beta = 1.5$ is squarely in the alpine hard-bed range, supporting the alpine component but not the low-relief or selective erosion components (at least not significantly). The weak segment-level positive $\Delta\beta$ (+0.27), if real, might reflect relict selective erosion structures visible at scales >50 km, but the current dataset cannot confirm this.

Highland A

Low- Relief / Selective erosion (relict)/ Alpine (subglacial)

Terminal

```
=== ASB_ICECAP_2010_Fig2G_Highland_A SUMMARY ===
-----
RESULTS SUMMARY (30 trajectories aggregated)
-----
VERTICAL RELIEF (Max-Min):
  -> Mean: 936.3m | Range: [142.2, 3247.2]m
ABSOLUTE BED ELEVATION:
  -> Min: -3445.5m | Max: 930.4m (across all trajectories)
AVG SEGMENT LENGTH:
  -> Mean: 113251.5m | Range: [15408.5, 413471.7]m
POWER LAW EXPONENT (Beta):
```

```

-> Mean: 2.0 | Range: [1.2, 3.1]
HURST EXPONENT:
-> Mean: 0.5 | Range: [0.1, 1.1]
MEAN ICE THICKNESS:
-> Mean: 3312.6m | Range: [2095.4, 3795.4]m
.....
CONFIRMED WAVELENGTHS (Physically valid < L/2):
-> Count: 2167
-> Mean: 578.6m | Range: [41.5, 32613.2]m
CANDIDATE WAVELENGTHS (Statistically present > L/2):
-> Count: 2
-> Range: [10983m, 38786m]
.....
=====
Exported 252 window rows and 35 segment rows

```

Observations

Roughness

The mean β value of 2.0 with a range: [1.2, 3.1] sits at the hard-bed/sedimentary boundary. The range spans almost 2 units, indicating substantial heterogeneity. The Hurst exponent $\zeta = 0.5$ (range [0.1, 1.1]) is intermediate, at the boundary between **anti-persistent and persistent behaviour**. This "boundary" character in both β and ζ suggests HA is a **transitional landscape** mixing hard-bed and sedimentary components.

Relief and bed elevation

The mean vertical relief of 936.3 m (range [142, 3247] m) is notable for its extreme variation. The minimum (142 m) represents near-flat terrain, while the maximum (3247 m) represents an extreme high relief. This extraordinary spread indicates some trajectories cross major subglacial mountain structures while others traverse low-relief basins.

The bed spans from -3445.5 m to +930.4 m. The -3445 m value represents an exceptionally deep subglacial trough. The total elevation range of ~4376 m is the largest observed yet among the data sets analysed.

Wavelength structure

The mean confirmed wavelength of 578.6 m (range [42 m, 32.6 km]) is consistent

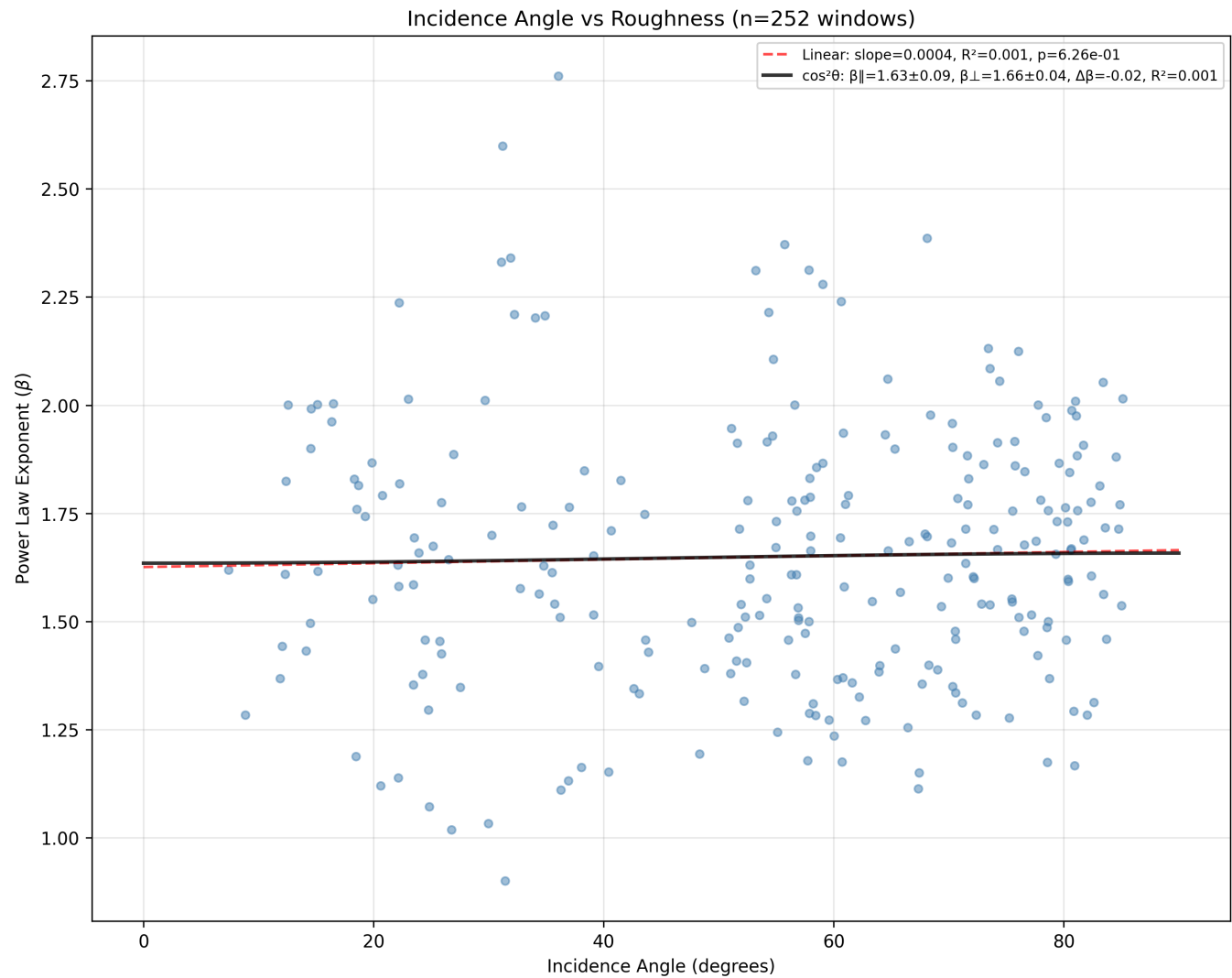
with **sub-km-scale roughness**. Only 2 candidate wavelengths exceeding $L/2$ are detected (at 11.0 km and 38.8 km). The sparse large-scale periodicity, despite enormous relief variation, suggests the topography is **irregular rather than periodic**. This would be consistent with alpine terrain where peaks and valleys are arranged without regular spacing, unlike the evenly spaced troughs of selective erosion.

The mean segment length of ~113 km (range [15, 413] km) and good sample size (30 trajectories, 252 windows, 35 segments) provide reasonable statistical power.

Mean ice thickness is 3312.6 m (range [2095, 3795] m).

Anisotropy

Window-level Roughness FIT



The window level analysis shows a linear fit with slope=0.0004, $R^2=0.001$, $p=6.26e-01$

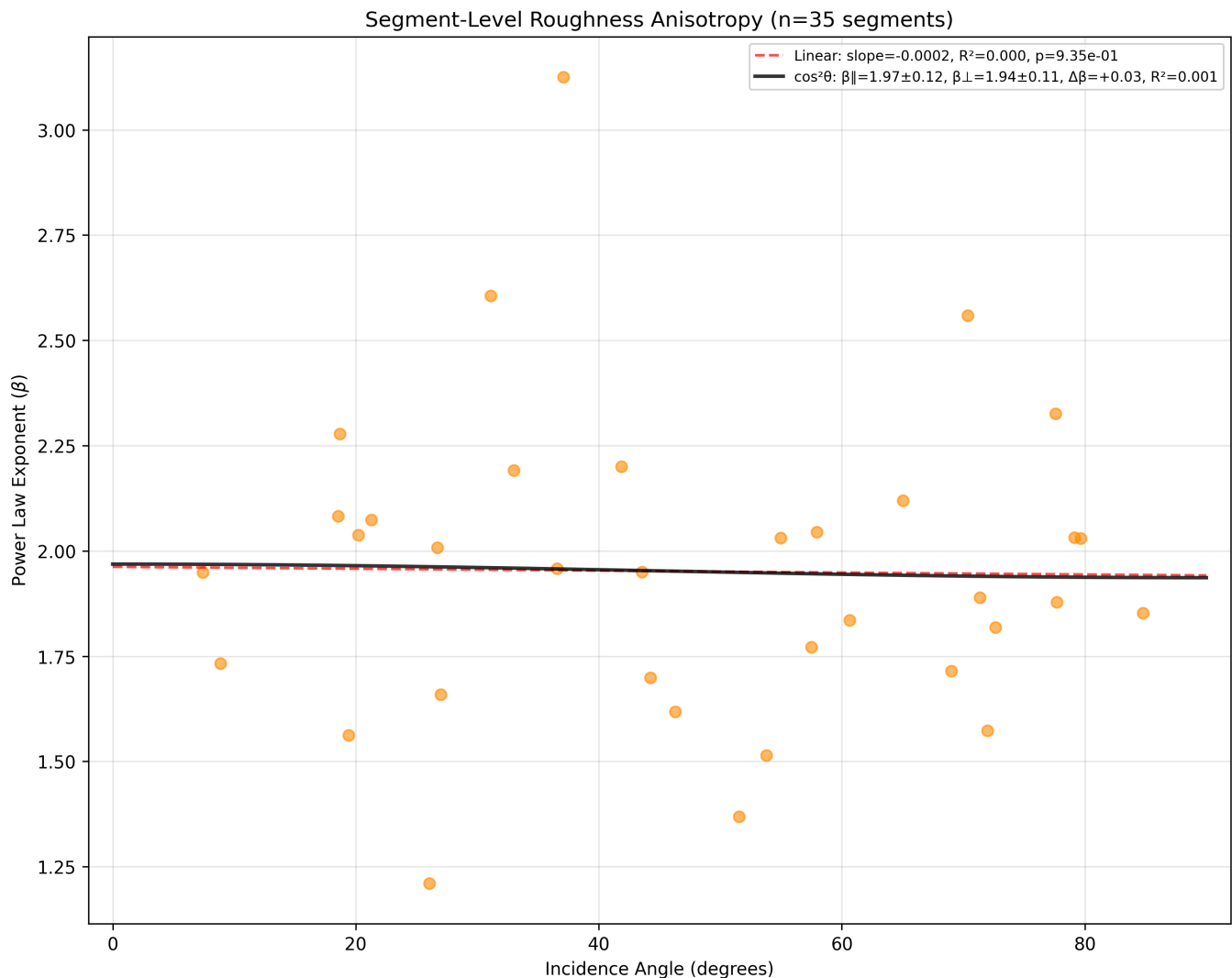
At $p=0.626$ this result is non-significant. Essentially **no directional dependence detected**.

The $\cos^2 \theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = -0.023$. Displays a weak signal for an **alpine landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.023}{\sqrt{0.086^2 + 0.045^2}} \approx \frac{0.023}{0.097} \approx 0.237$

This is below 1.96 (95% confidence interval), which means that the result is not significant. Since the dataset is small and the $R^2=0.0006$ implies that the model explains only 0.06% of variance. This means that **essentially no directional dependence is detected by the model**. The result is isotropic.

Segment-level Roughness FIT



The segment level analysis shows a linear fit with slope=-0.0002, $R^2=0.000$, $p=9.35e-01$

At $p=0.935$ this result is also non-significant.

The $\cos^2 \theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = +0.032$, which provides support for the **selective erosion landscape**.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.032}{\sqrt{0.118^2 + 0.108^2}} \approx \frac{0.032}{0.160} \approx 0.200$

This is below 1.96 (95% confidence interval), which means that the result is not significant. $R^2=0.0009$ implies that the model explains 9% of variance.

Note: the segment-level β values ($\beta_{\parallel} = 1.97$, $\beta_{\perp} = 1.94$) are noticeably higher than window-level ($\beta_{\parallel} = 1.63$, $\beta_{\perp} = 1.66$). This scale-dependent β increase suggests that when averaging over longer segments, the smoother/sedimentary components gain influence, while the 50 km windows capture more of the local hard-bed roughness.

Conclusion:

HA shows **extreme topographic heterogeneity** (relief range [142, 3247] m) with the deepest bed elevation of any region (-3446 m), yet the roughness anisotropy is **clearly isotropic** at both scales (values are essentially zero). This is the classic signature of an **alpine landscape** (irregular, deeply dissected terrain with no preferred structural orientation).

The IFPA classification of "Low-Relief / Selective erosion (relict) / Alpine (subglacial)" is partially supported: the low window-level β (~ 1.6) and clear isotropy are consistent with the alpine component. The low-relief component is present in the minimum relief trajectory (142 m), but the mean relief (936 m) and extreme maximum (3247 m) suggest alpine terrain dominates. The selective erosion component is **not supported** by the anisotropy analysis (no directional signal is detected at either scale).

The sparse candidate wavelengths (only 2) further support the alpine interpretation: the terrain lacks the regular periodicity expected from selective erosion and is instead irregularly structured.

Golicyna Subglacial Highlands

Low relief/ Selective erosion

Terminal

=== ASB_ICECAP_2010_Fig2H_Golicyna_SH SUMMARY ===

RESULTS SUMMARY (19 trajectories aggregated)

VERTICAL RELIEF (Max-Min):

-> Mean: 699.5m | Range: [125.0, 1868.0]m

ABSOLUTE BED ELEVATION:

-> Min: -1453.6m | Max: 634.7m (across all trajectories)

AVG SEGMENT LENGTH:

-> Mean: 133226.2m | Range: [19603.3, 342414.5]m

POWER LAW EXPONENT (Beta):

-> Mean: 1.8 | Range: [1.1, 2.6]

HURST EXPONENT:

-> Mean: 0.4 | Range: [0.1, 0.8]

MEAN ICE THICKNESS:

-> Mean: 3533.1m | Range: [2987.3, 4132.1]m

.....
CONFIRMED WAVELENGTHS (Physically valid < L/2):

-> Count: 2336

-> Mean: 497.6m | Range: [35.9, 31510.2]m

CANDIDATE WAVELENGTHS (Statistically present > L/2):

-> Count: 13

-> Range: [8599m, 31741m]
.....

=====

Exported 249 window rows and 39 segment rows

Observations

Roughness

The mean $\beta = 1.8$ (range [1.1, 2.6]) is in the hard-bed regime. The range is moderate (1.5 unit spread). The Hurst exponent $\zeta = 0.4$ (range [0.1, 0.8]) is low, indicating **anti-persistent (rough, reversing) topography**. The combination of $\beta = 1.8$ with $\zeta = 0.4$ indicates a predominantly hard-bed landscape with crystalline bedrock creating high-frequency roughness.

Relief and bed elevation

The mean vertical relief of 699.5 m (range [125, 1868] m) is moderate with substantial variation (~15x ratio). Some trajectories show very low relief (125 m),

while others reach 1868 m. This spread suggests a mix of subdued highland terrain and more deeply dissected areas.

The bed spans from -1453.6 m to +634.7 m. The modest maximum elevation (+635 m) is the lowest of any region, which is surprising for a region named "Subglacial Highlands" and suggests these are relatively subdued features compared to HD or HA.

Wavelength structure

The mean confirmed wavelength of 497.6 m (range [36 m, 31.5 km]) is the lowest of all regions, indicating roughness dominated by the most closely spaced features. This is consistent with the hard-bed interpretation (crystalline bedrock creates dense, high-frequency roughness).

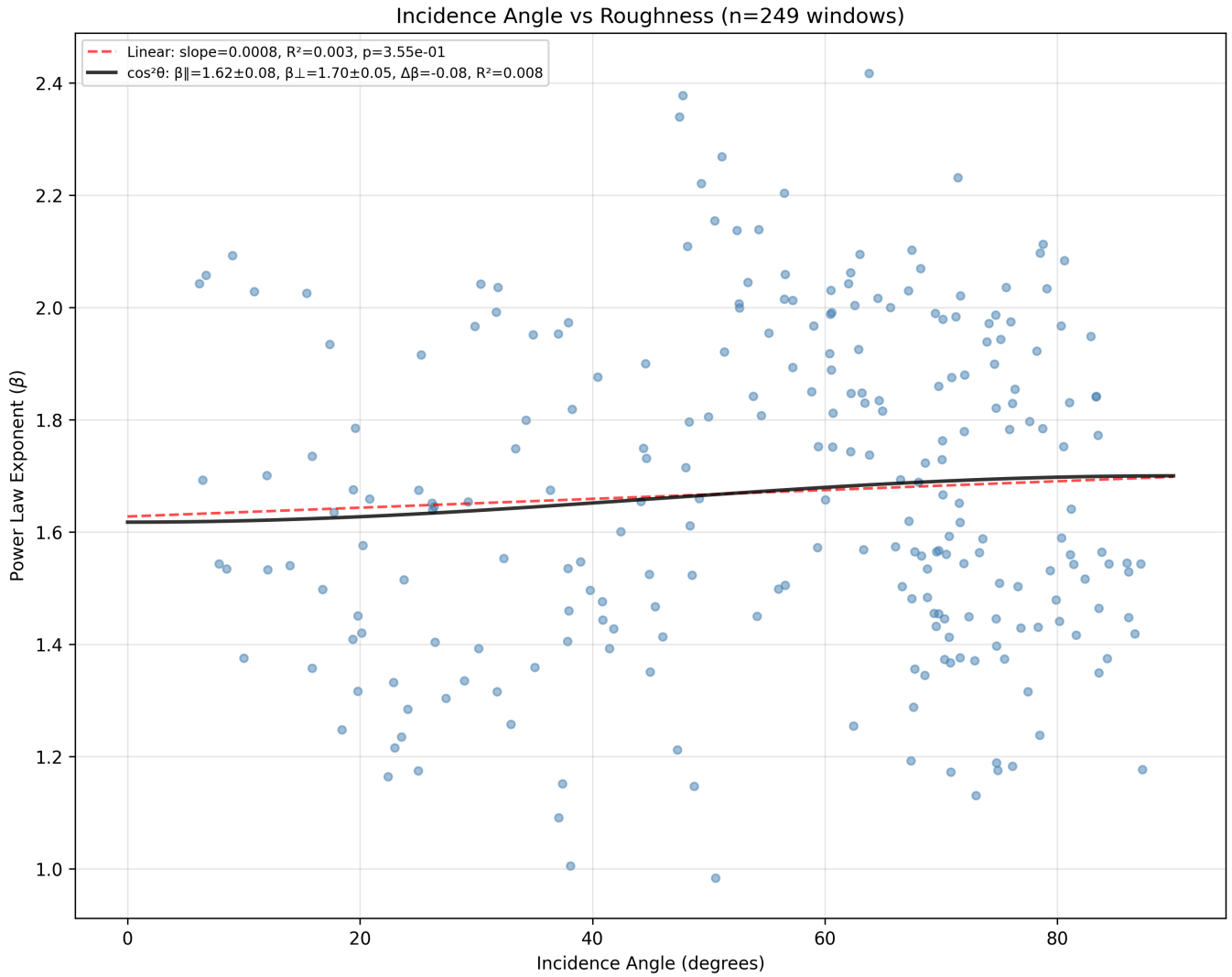
Thirteen candidate wavelengths exceeding $L/2$ are detected, spanning 8.6–31.7 km. Combined with the low β , this suggests the terrain has both fine-scale crystalline roughness *and* significant large-scale periodic structures, possibly relict valley/ridge systems from pre-glacial drainage or ice-stream modification.

The mean segment length of ~133 km (range [20, 342] km) provides excellent resolution for detecting large-scale features. The good sample size (19 trajectories, 249 windows, 39 segments) supports reasonable statistical power.

Mean ice thickness is 3533.1 m (range [2987, 4132] m) reflects the deep interior location.

Anisotropy

Window-level Roughness FIT



The window level analysis shows a linear fit with slope=0.0008, $R^2=0.003$, $p=3.55e-01$

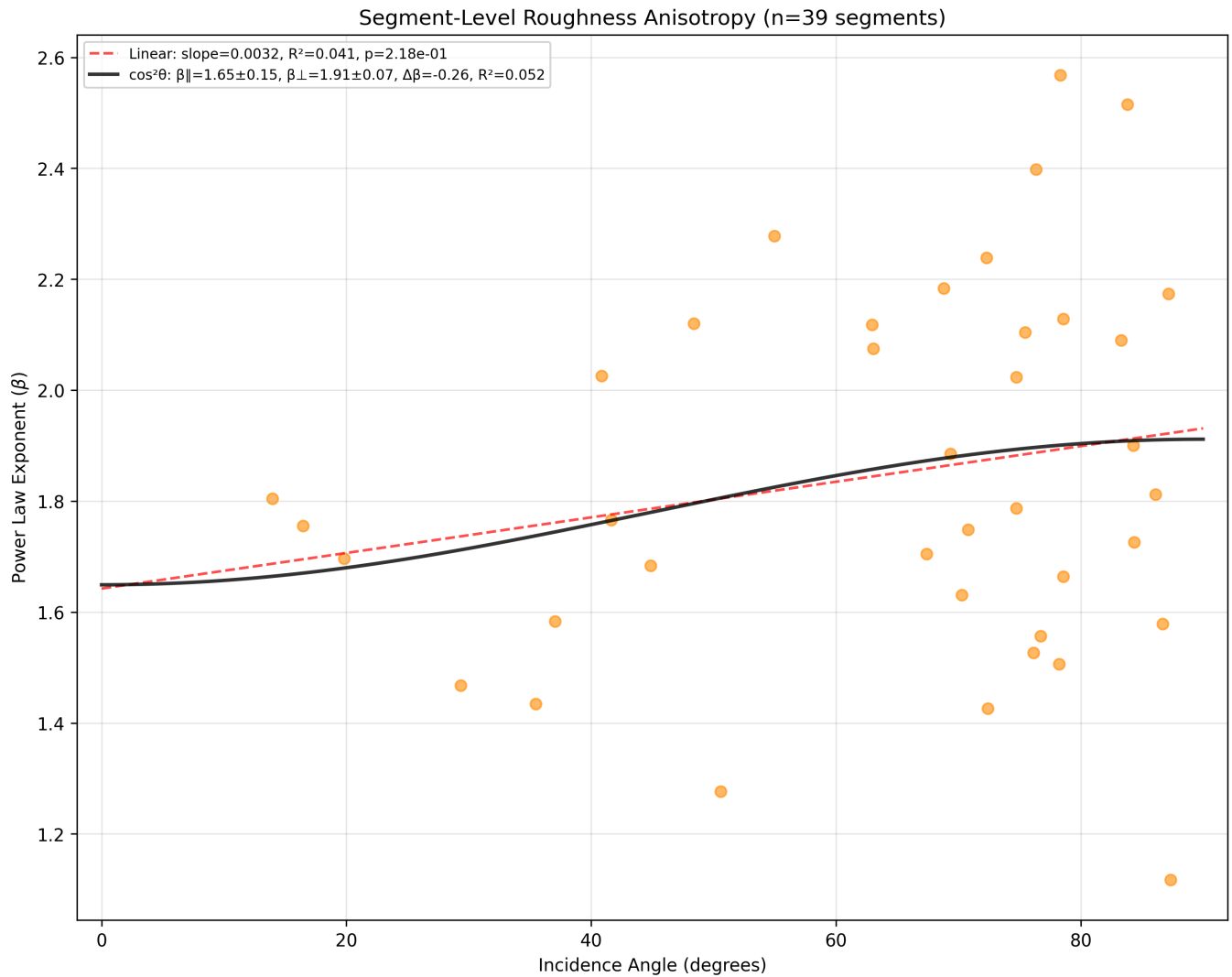
At $p=0.355$ this result is non-significant.

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{||} - \beta_{\perp} = -0.083$, which provides support for the **Alpine landscape** class.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{||}^2 + \sigma_{\perp}^2}} = \frac{0.083}{\sqrt{0.080^2 + 0.045^2}} \approx \frac{0.083}{0.092} \approx 0.904$

This is below 1.96 (95% confidence interval), which means that the result is not significant. Since the dataset is small and the $R^2=0.0079$ implies that the model explains 0.79% of variance. This is a **weak alpine signal** (suggestive but not conclusive).

Segment-level Roughness FIT



The segment level analysis shows a linear fit with slope=0.0032, $R^2=0.041$, $p=2.18e-01$

At $p=0.218$ this result is non-significant.

The $\cos^2\theta$ anisotropy signal is $\Delta\beta = \beta_{\parallel} - \beta_{\perp} = -0.262$, which provides support for the **Alpine landscape**. The signal grows from -0.083 to -0.262, and both scales agree in sign.

The fit z-score can be calculated by: $\frac{|\Delta\beta|}{\sqrt{\sigma_{\parallel}^2 + \sigma_{\perp}^2}} = \frac{0.262}{\sqrt{0.150^2 + 0.069^2}} \approx \frac{0.262}{0.165} \approx 1.587$

This is below 1.96 (95% confidence interval), which means that the result is not significant. $R^2=0.0519$ implies that the model explains 5% of variance. The strengthening with scale suggests that in GSH, the alpine ridge/valley structures may be larger-scale features (~100+ km) that are better captured by segment-level averaging than by individual 50 km windows.

Conclusion:

GSH shows a **hard-bed landscape** ($\beta = 1.8$, $\zeta = 0.4$) with the lowest mean wavelength (498 m) and highest ice thickness (3533 m) of all regions. The anisotropy analysis reveals a **consistent negative $\Delta\beta$ at both scales** (weak at window level but strengthening to $z = 1.59$ at segment level) suggesting a **probable alpine signature** that falls just short of statistical significance. More data might provide higher certainty in this result.

The IFPA classification of "Low relief / Selective erosion" is **only partially supported**. The low β confirms hard bed (consistent with crystalline highlands), but the negative $\Delta\beta$ (indicating alpine rather than selective erosion anisotropy) contradicts the selective erosion component. The 13 candidate wavelengths at 9–32 km indicate significant periodic structure, which could represent relict drainage patterns rather than active ice-stream erosion.

The relatively low maximum bed elevation (+635 m) suggests these "highlands" are modest features, possibly deeply buried under thick ice (mean 3533 m). The name "Subglacial Highlands" may reflect relative elevation within the basin rather than dramatic topographic prominence.